

Named Entity Recognition and Disambiguation

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**Politecnico
di Torino**

Lecture goal

- Definition of NERD
- Application contexts
- Rule-based NER
- Knowledge-based NER
- Machine Learning-based NER
- Transformer-based NER

Preliminary example

Person p Loc l Org o Event e Date d Other z

Barack Hussein Obama II * (born August 4, 1961 *) is an American * attorney and politician who served as the 44th President of the United States * from January 20, 2009 *, to January 20, 2017 *. A member of the Democratic Party *, he was the first African American * to serve as president. He was previously a United States Senator * from Illinois * and a member of the Illinois State Senate *.

Outline

- Recognition
 - Recognize *Barack Hussain Obama II*
- Disambiguation
 - Distinguish between *Cars (movie)* and *Cars (vehicles)*
- Resolution
 - Fix typos (e.g., *Obama Barackk*)
 - reconduct to a normal form (e.g., *Barack Hussain Obama II*)

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70%

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- 20%

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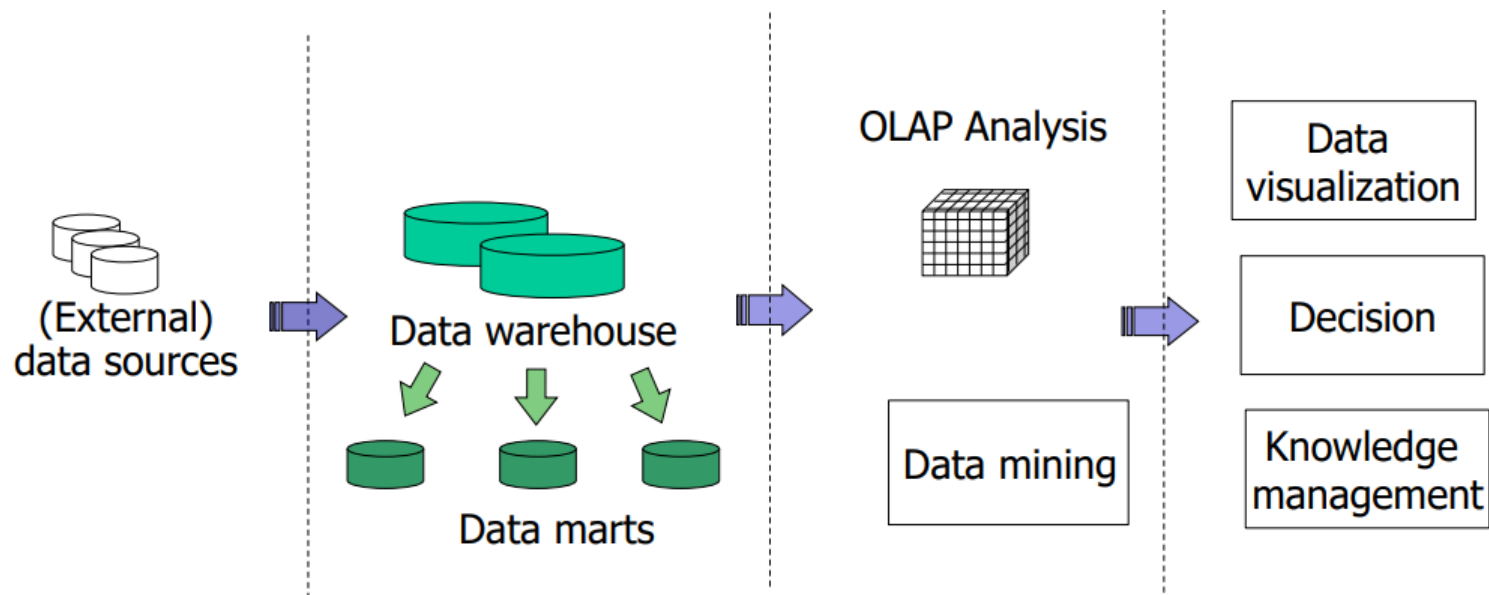
10%

Goal

- Find semantic references to specific entities of the real world
- Applications
 - Business intelligence
 - Information retrieval
 - Text classification and intent detection
 - Question answering and chatbotting
 - Summarization

Business Intelligence

- Provide support in companies to make strategic decisions
- transform company data into actionable information



Data warehouse introduction. Elena Baralis. Politecnico di Torino. https://dbdmg.polito.it/dbdmg_web/index.php/2021/09/13/data-management-and-visualization-2021-22/

Information Retrieval

- Software systems that provides access to document collections
 - Full-text search
 - Search based on metadata

<https://it-it.facebook.com> > ... > Intesa Sanpaolo ▾
Intesa Sanpaolo - Home | Facebook
La pagina Facebook di **Intesa Sanpaolo** è uno spazio pensato per chiedere informazioni e ricevere assistenza sui prodotti, i servizi e le iniziative della ...

<http://www.intesasnanpaoloprivatebanking.it> ▾
Intesa Sanpaolo Private Banking
... Assistenza clienti · Obbligazioni e certificati · Carte: conversione valutaria. © 2016 **INTESA SANPAOLO PRIVATE BANKING** | Tutti i diritti riservati.

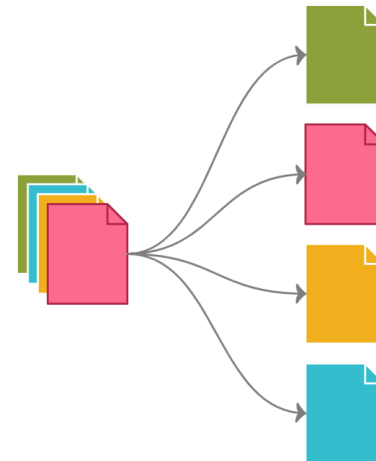
<https://it.wikipedia.org> > wiki > Intesa_Sanpaolo ▾
Intesa Sanpaolo - Wikipedia
Intesa Sanpaolo è un istituto bancario italiano attivo dal 1° gennaio 2007, nato dalla fusione tra Banca Intesa e Sanpaolo IMI.
Dipendenti: 91.478 (2019) Fondata da: [Banca Intesa](#); [Sanpaolo IMI](#)
Fatturato: 18,08 miliardi di € (2019) Stato: [Italia](#)

<https://www.linkedin.com> > company > intesa-sanpaolo ▾
Intesa Sanpaolo | LinkedIn
Intesa Sanpaolo | 401190 followers on LinkedIn. **Intesa Sanpaolo**, the leading Italian banking group, offers its services to 11.1 million customers through a ...

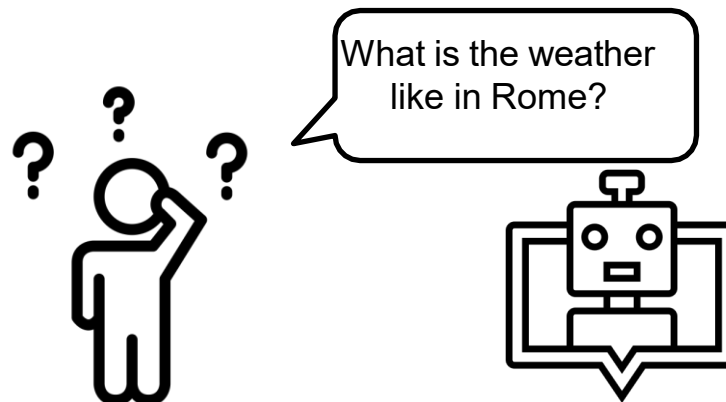
<https://www.google.it/search?q=intesa+sanpaolo>

Text classification

- Classify unlabeled pieces of text into one or more predefined classes
 - It requires the availability of a set of annotated documents, i.e., the training set
- Applications
 - Sentiment analysis
 - Topic discovery
 - Language identification
 - Hate speech detection
 - Intent detection



Intent detection



Icons credit: Creative Mania & Rahmat Dwi Cahyo from the Noun Project

Intent detection

- The system is unaware of the action users are willing to take
- NER and NED modules are used to understand the entities involved



1

NER module

Rome: LOC

2

NED module

Rome: /wiki/Rome

Question answering

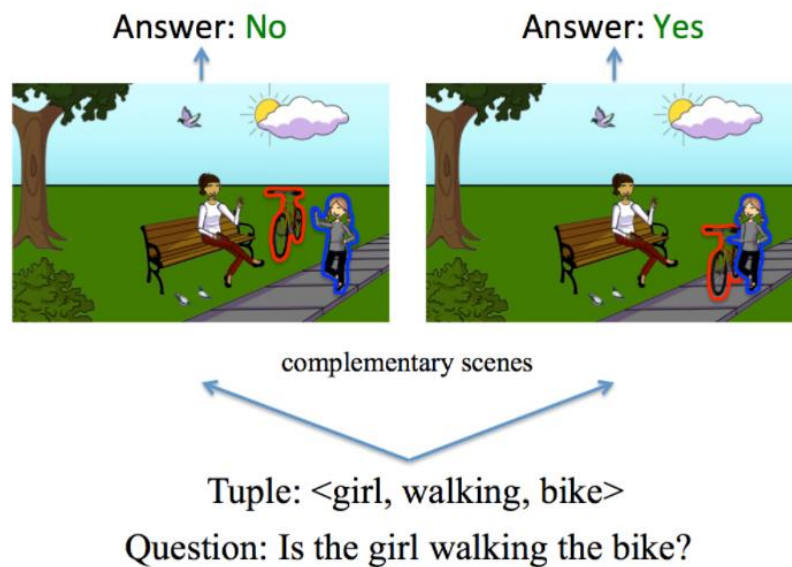
- Provide relevant answers in response to questions posed in natural language
- Main steps
 - Question classification
 - Information Retrieval
 - Answer extraction



flyelite.com (latest access: April 2021)

Question answering

- Not only text...



<https://visualqa.org/> (latest access: April 2021)

Summarization

- Produce a concise version of a (multimodal) collection that incorporates the most salient content

Transcript

today we are going to show you how to make spanish omelet . i 'm going to dice a little bit of peppers here . i 'm not going to use a lot , i 'm going to use very very little . a little bit more then this maybe . you can use red peppers if you like to get a little bit color in your omelet . some people do and some people do n't t is the way they make there spanish omelets that is what she says . i loved it , it actually tasted really good . you are going to take the onion also and dice it really small . you do n't want big chunks of onion in there cause it is just pops out of the omelet . so we are going to dice the up also very very small . so we have small pieces of onions and peppers ready to go .



Summary

how to cut peppers to make a spanish omelette; get expert tips and advice on making cuban breakfast recipes in this free cooking video .

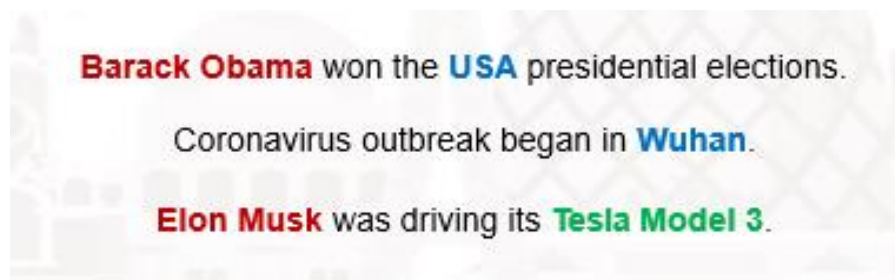
Multimodal Abstractive Summarization for How2 Videos. Shruti Palaskar, Jindrinch Libovicky, Spandana Gella, Florian Metze. <https://arxiv.org/abs/1906.07901>

Entity Recognition

- The main purpose is to recognize the entities
- **Entities** consist of any word/multiword forms in a piece of text that refers to **identifiable concepts**
 - E.g., text: *Book a movie ticket*
 - “Movie” is a candidate entity

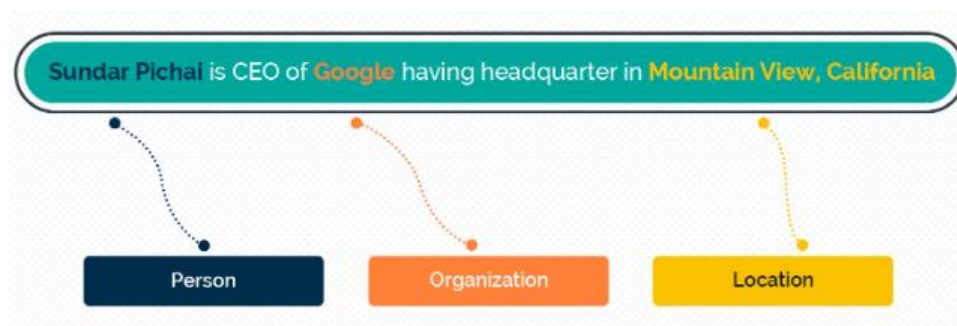
Named Entities

- Named entities are real-world objects, persons, locations, etc..
- Unlike other entity types, they link to a specific concept in the real world

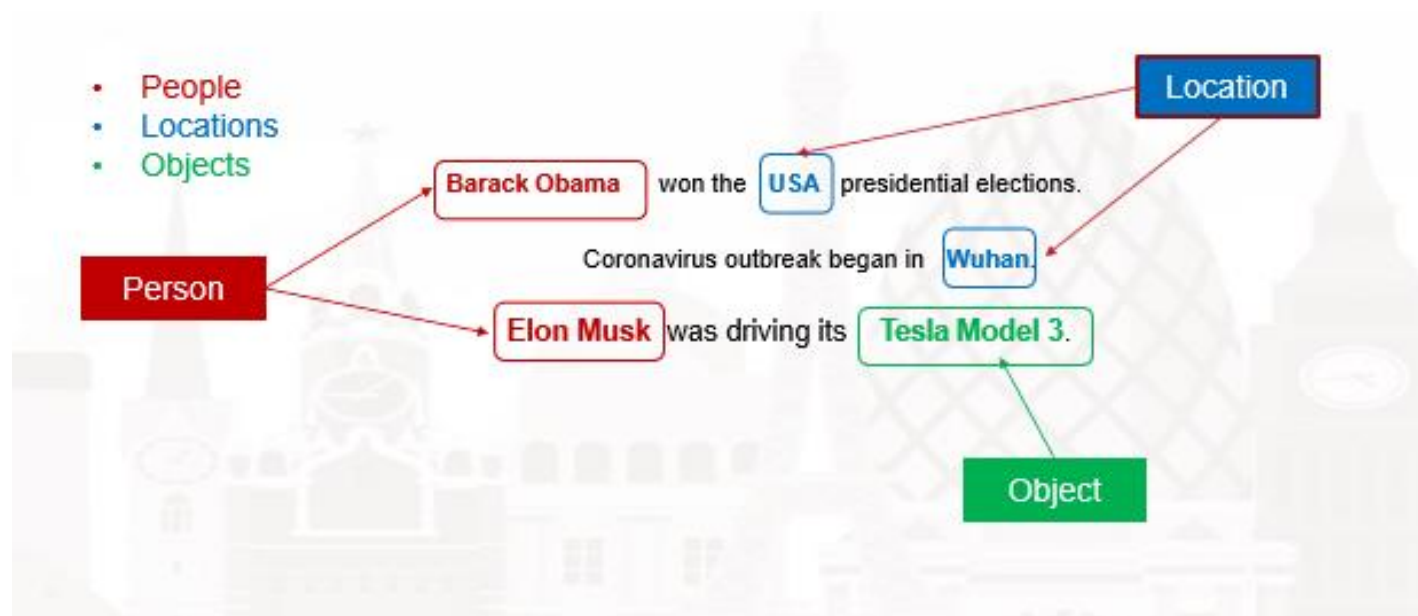


Named Entity Recognition

- Identify the named entities in the text
 - Define the corresponding text span
- Classify the detected entities in one of the predefined set of classes
 - Assign a category (classification)



Entity Recognition: example



Entity recognition – Spacy NER entities

PERSON: People, including fictional.
NORP: Nationalities or religious or political groups.
FAC: Buildings, airports, highways, bridges, etc.
ORG: Companies, agencies, institutions, etc.
GPE: Countries, cities, states.
LOC: Non-GPE locations, mountain ranges, bodies of water.
PRODUCT: Objects, vehicles, foods, etc. (Not services.)
EVENT: Named hurricanes, battles, wars, sports events, etc.
WORK_OF_ART: Titles of books, songs, etc.
LAW: Named documents made into laws.
LANGUAGE: Any named language.
DATE: Absolute or relative dates or periods.
TIME: Times smaller than a day.
PERCENT: Percentage, including "%".
MONEY: Monetary values, including unit.
QUANTITY: Measurements, as of weight or distance.
ORDINAL: "first", "second", etc.
CARDINAL: Numerals that do not fall under another type.

Entity Recognition – Spacy NER

- Try to guess the recognized entities...

The Polytechnic University of Turin (Italian: Politecnico di Torino) is a public university based in Turin, Italy.

Established in 1859, it is Italy's oldest technical university.

The university offers several courses in the fields of Engineering, Architecture and Industrial Design.

Entity Recognition –Spacy NER

- Good but not great...

The Polytechnic University of Turin ORG (Italian NORP : Politecnico di Torino GPE) is a public university based in Turin GPE , Italy GPE . Established in 1859, it is Italy GPE 's oldest technical university. The university offers several courses in the fields of Engineering, Architecture and Industrial Design ORG .

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Missing ORG entity

Entity Recognition –Spacy NER

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Missing DATE entity

Entity Recognition –Spacy NER

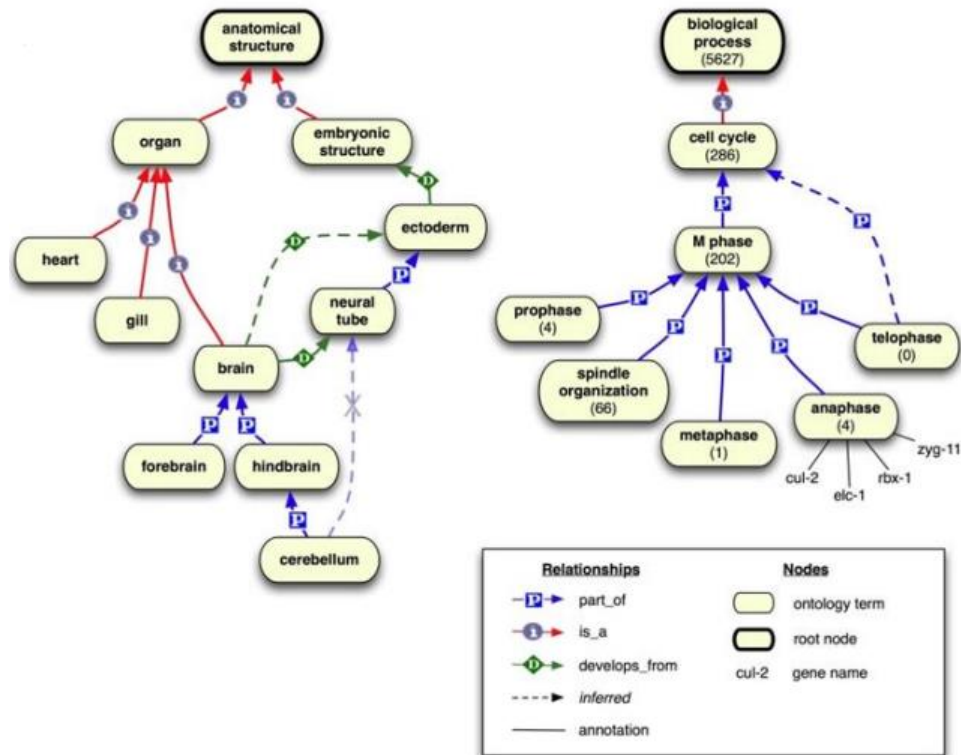
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Correct label, but
wrong text span

Semantic models

- Entities are usually stored in (domain-specific) semantic models called **ontologies**

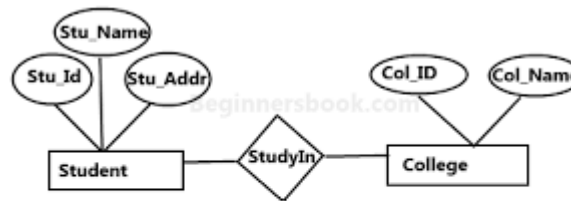


Semantics in the ICT domain

- Common meaning in semantic technology helps computer systems more accurately interpret what people mean.
- Common meaning enables disparate IT systems – data sources and applications – to interface more efficiently and productively

Semantic data models

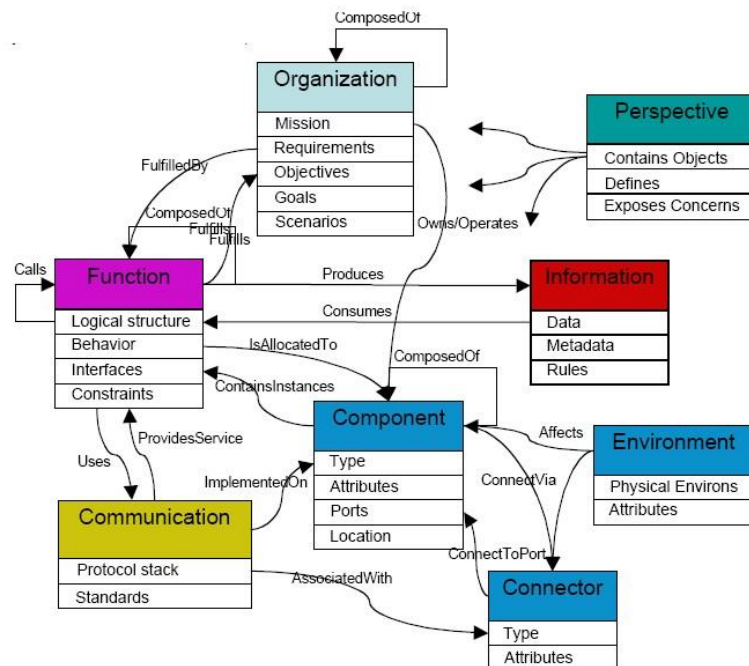
- Semantic data models (SDMs, in short) are high-level semantics-based database descriptions and structuring formalism (database model) for databases
- They describes a database in terms of
 - the kinds of entities that exist in the application environment
 - the classifications and groupings of those entities,
 - the structural interconnections among them



Sample E-R Diagram

Why do we need semantic models?

- Semantic models like vocabularies, taxonomies, and ontologies explicitly model semantic relationships
- They are the pillar of traditional NLP approaches
 - Commonly used before the advent of Deep NLP
 - Still used in Deep NLP for addressing particular tasks

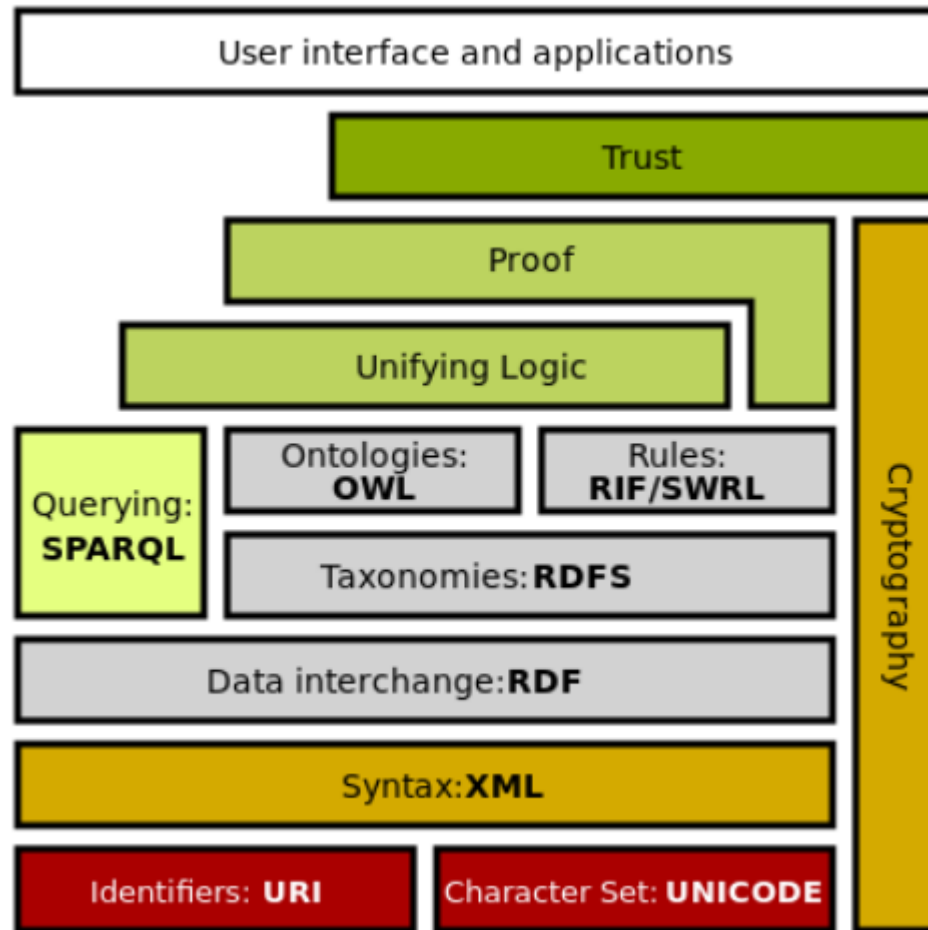


The Semantic Web

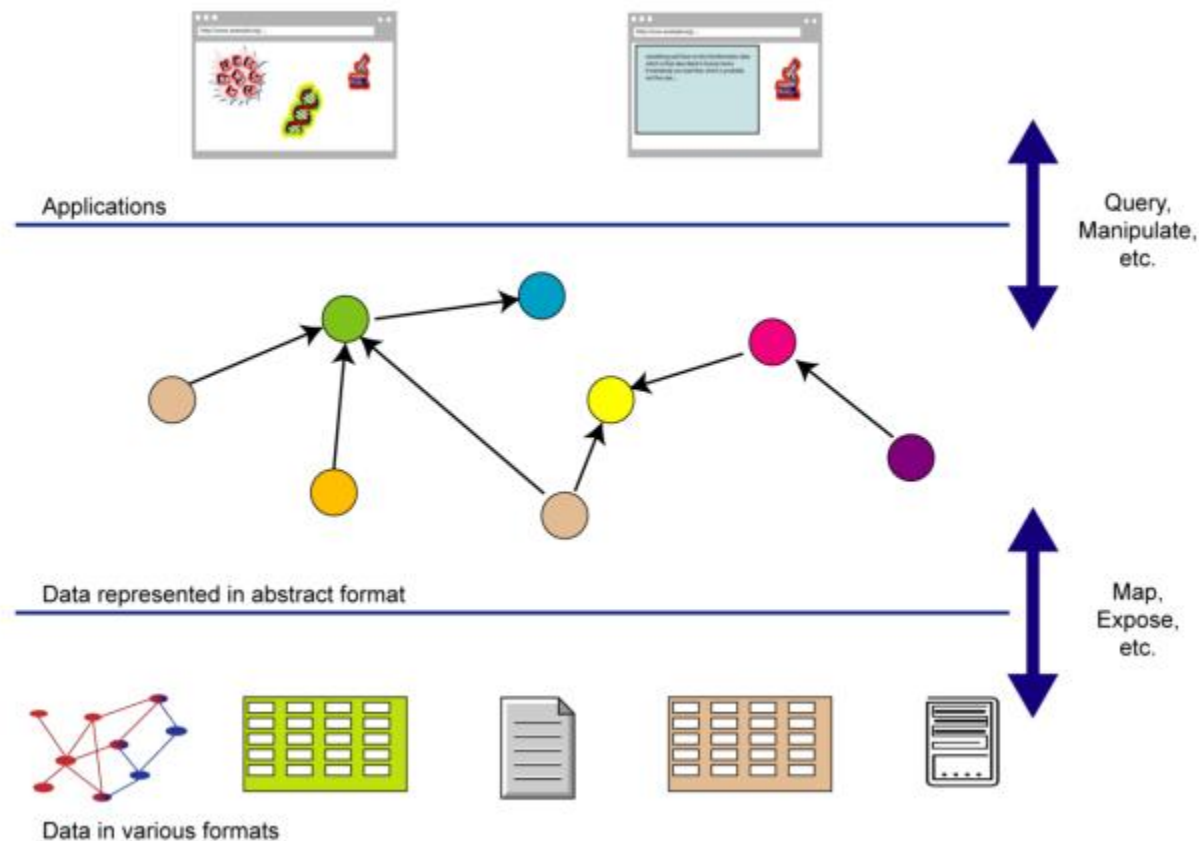
- Extension of the World Wide Web through standards set by the World Wide Web Consortium (W3C)
- Stack of enabling technologies
- Goal: make Internet data machine-readable



The Semantic Web



The Semantic Web



Resource Description Framework (RDF)

- A Resource corresponds to a Uniform Resource Identifier
- A RDF is structured in statements
- A statement is a triple <Subject – Predicate – Object>
 - Subject: a resource
 - Predicate: a verb / property / relationship
 - Object: a resource, or a literal string
- RDF/XML
 - XML is a syntax
 - RDF is a data model

Resource Description Framework

- RDF is a framework for describing resources on the web
 - E.g., purchased items, documents, pictures, videos
- It is designed to be read and understood by computers
- It is not designed for being displayed to people
- It is usually written in XML

```
<?xml version="1.0"?>

<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:si="https://www.w3schools.com/rdf/">

  <rdf:Description rdf:about="https://www.w3schools.com">
    <si:title>W3Schools</si:title>
    <si:author>Jan Egil Refsnes</si:author>
  </rdf:Description>

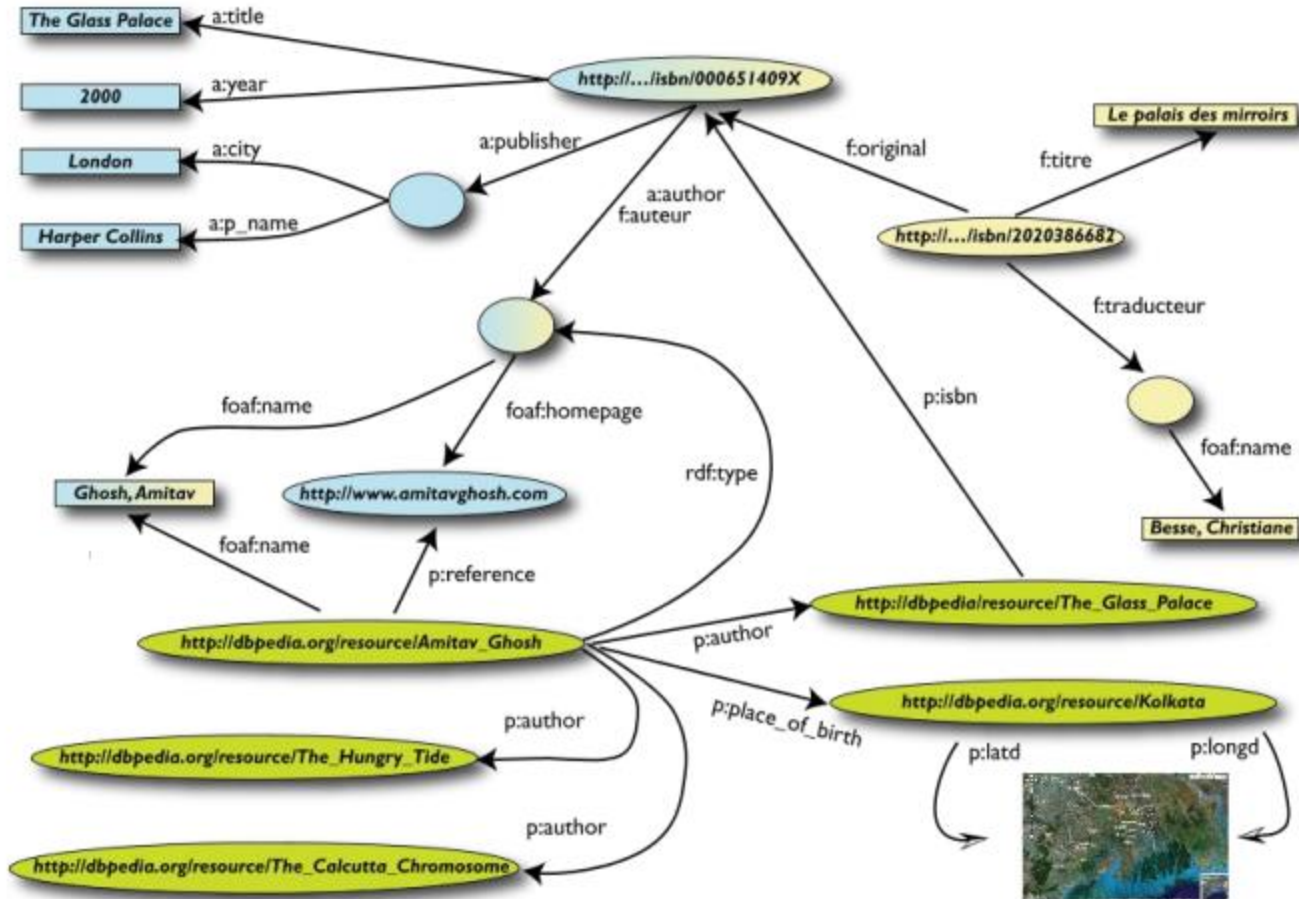
</rdf:RDF>
```

<https://www.w3schools.com/> Latest access: July 2021

Resource Description Framework

- Examples of use
 - Describing properties for shopping items, such as price and availability
 - Describing time schedules for web events
 - Describing information about web pages (content, author, created and modified date)
- Describing content and rating for web pictures
- Describing content for search engines
- Describing electronic libraries

RDF example



Ontologies

- Ontologies aim at describing the world without semantic ambiguity
- are used for automatic reasoning as their structure is machine-readable
- Key elements (not exhaustive list)
 - **Classes or concepts** are the nodes of the graph
 - E.g., “portrait” or “painter”
 - **Individuals** are instances of abstract concepts
 - E.g., “The Mona Lisa” or “Leonardo Da Vinci”
 - **Relations** are edges of the graph. They are used to link concepts
 - **Links to descriptors**
 - E.g., dcterms:title links the object with its title

Ontology components (detailed list)

- Individuals
 - instances or objects (the basic or "ground level" objects)
- Classes
 - sets, collections, concepts, types of objects, or kinds of things
- Attributes
 - aspects, properties, features, characteristics, or parameters that objects (and classes) can have
- Relations
 - ways in which classes and individuals can be related to one another
- Events
 - the changing of attributes or relations

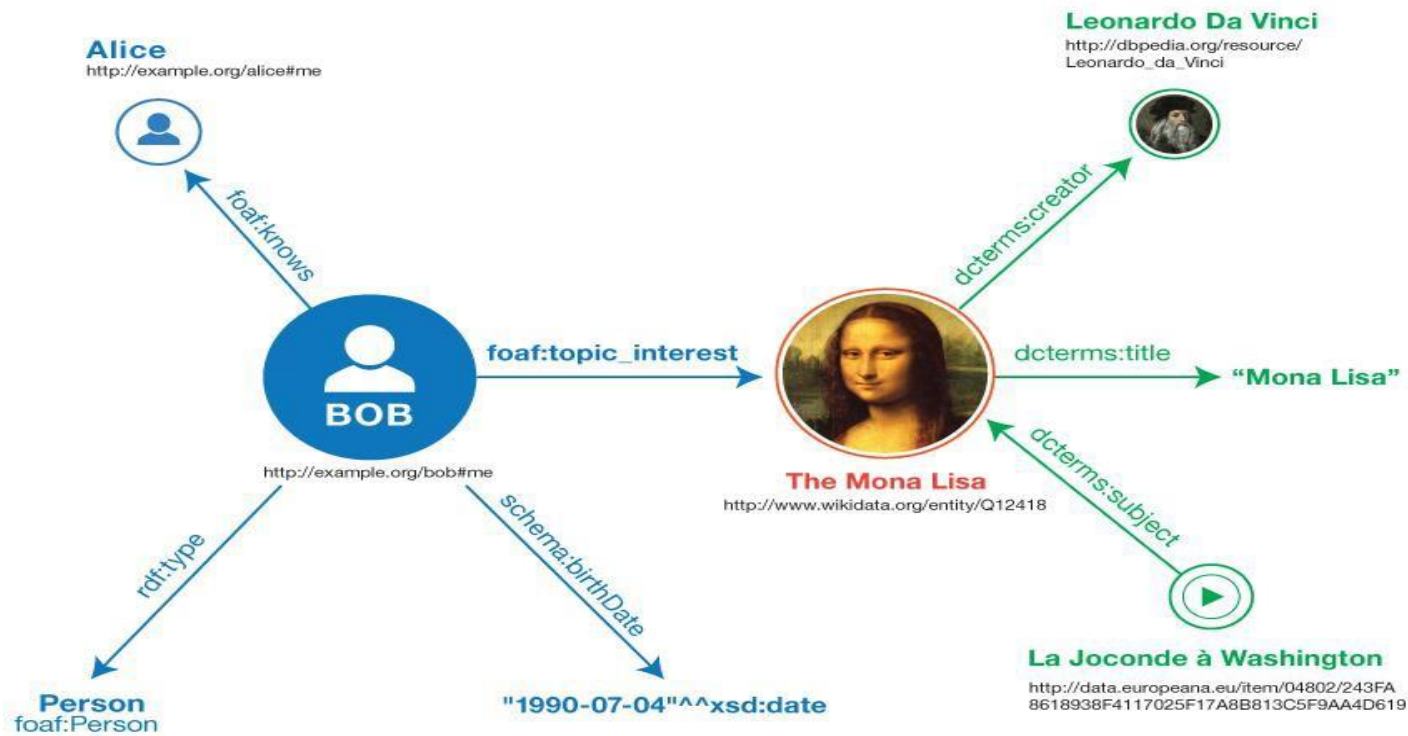
Maureen Donnelly; Giancarlo Guizzardi (2012). Formal Ontology in Information Systems: Proceedings of the Seventh International Conference (FOIS 2012). IOS Press. ISBN 978-1-61499-083-3.

Ontology components (detailed list)

- Function terms
 - complex structures formed from certain relations that can be used in place of an individual term in a statement
- Restrictions
 - formally stated descriptions of what must be true in order for some assertion to be accepted as input.
- Rules
 - statements in the form of an if-then (antecedent-consequent) sentence that describe the logical inferences that can be drawn from an assertion in a particular form.
- Axioms
 - assertions (including rules) in a logical form that together comprise the overall theory that the ontology describes in its domain of application.
- Events
 - the changing of attributes or relations

Maureen Donnelly; Giancarlo Guizzardi (2012). Formal Ontology in Information Systems: Proceedings of the Seventh International Conference (FOIS 2012). IOS Press. ISBN 978-1-61499-083-3.

Ontologies

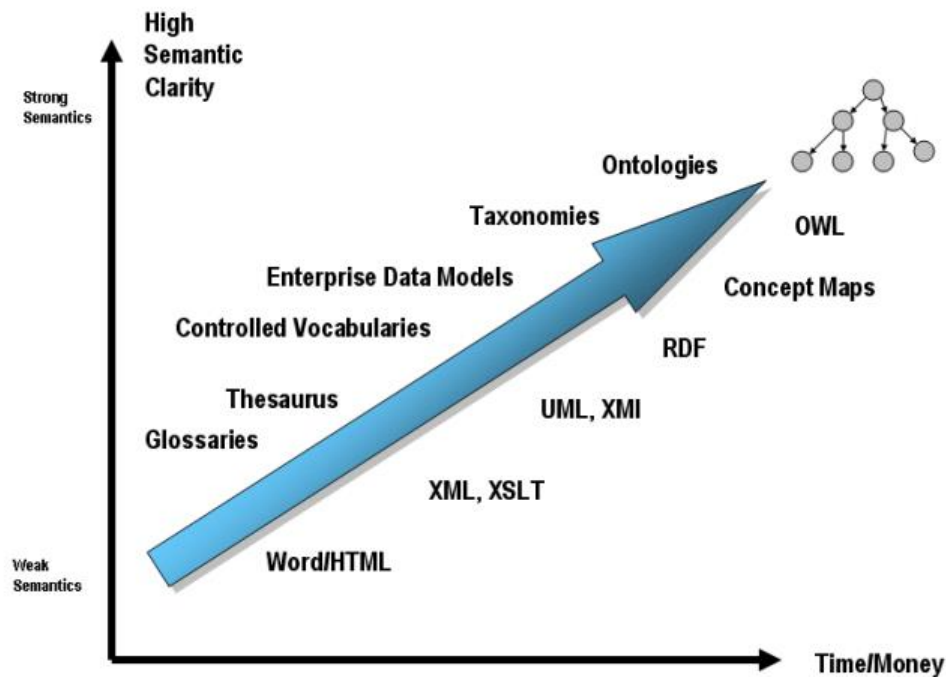


Do we always need ontologies for NER?

- Probably not
 - We often need to model covering just a subject of real-world objects and relations
 - We can simplify/generalize the relations between ontology components
 - We do not always need ad hoc search tools
 - Querying a complete ontology can be costly

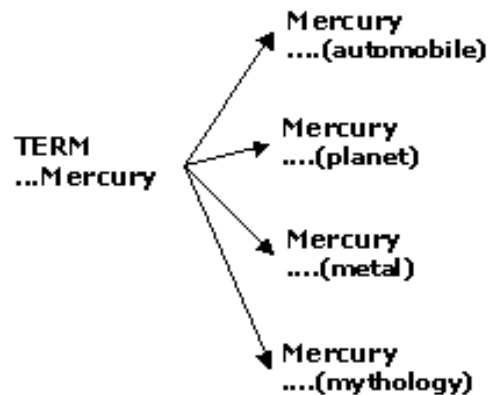
What if ontologies are not available or not necessary?

- Simpler semantic models are used



Controlled vocabulary

- List of terms of phrases
 - E.g., indices, subject headings
- Common usage
 - Describe semantically relevant concepts and relationships
 - Prevent linguistic errors and ambiguities in natural language
 - Gather linguistic relations



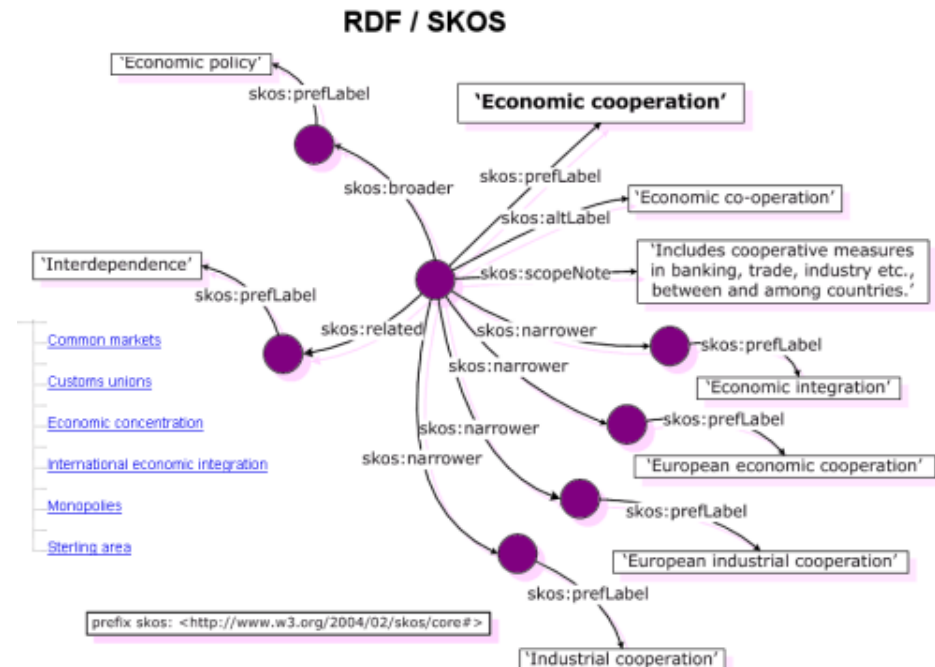
Taxonomy

- Hierarchical representation of concepts within a controlled vocabulary
- Specific for a given subject
- Relationships
 - Parent = broader term
 - Child = narrower term



Thesauri

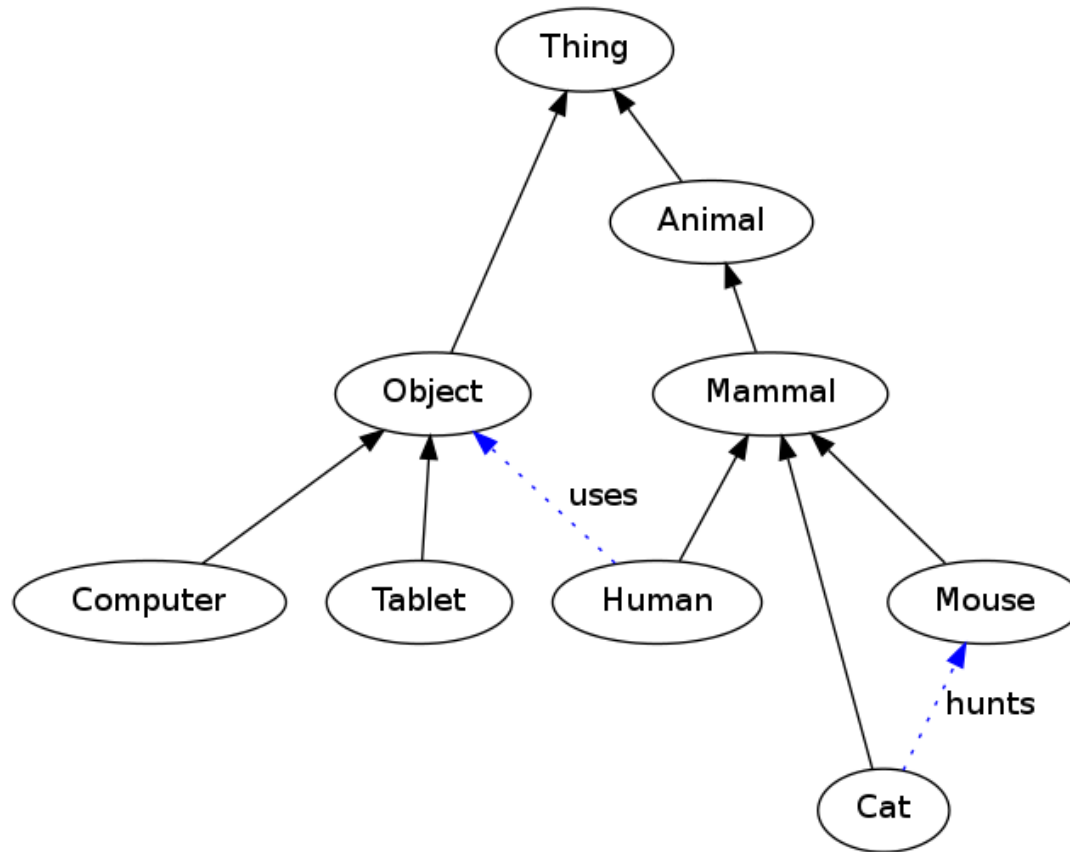
- Extended taxonomy models
- Specific for a given subject
- Relationships
 - Parent = broader term
 - Child = narrower term



Knowledge Graphs

- Also denoted by Semantic Networks
- networks of real-world entities
- Usually stored in a graph database
- Made up of three main components: nodes, edges, and labels
- Graph nodes are objects, events, situations, or concepts
- Graph edges illustrate the relationship between entities

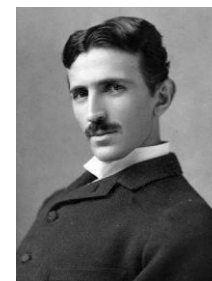
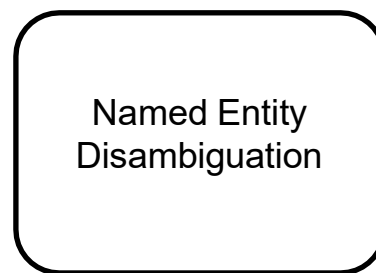
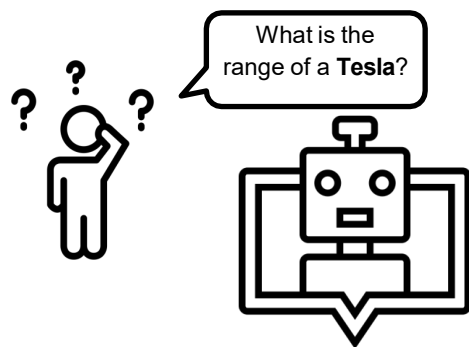
KG example



Harispe, S., Ranwez, S., Janaqi, S., & Montmain, J. (2017). Semantic Measures for the Comparison of Units of Language, Concepts or Entities from Text and Knowledge Base Analysis.

Named Entity Disambiguation

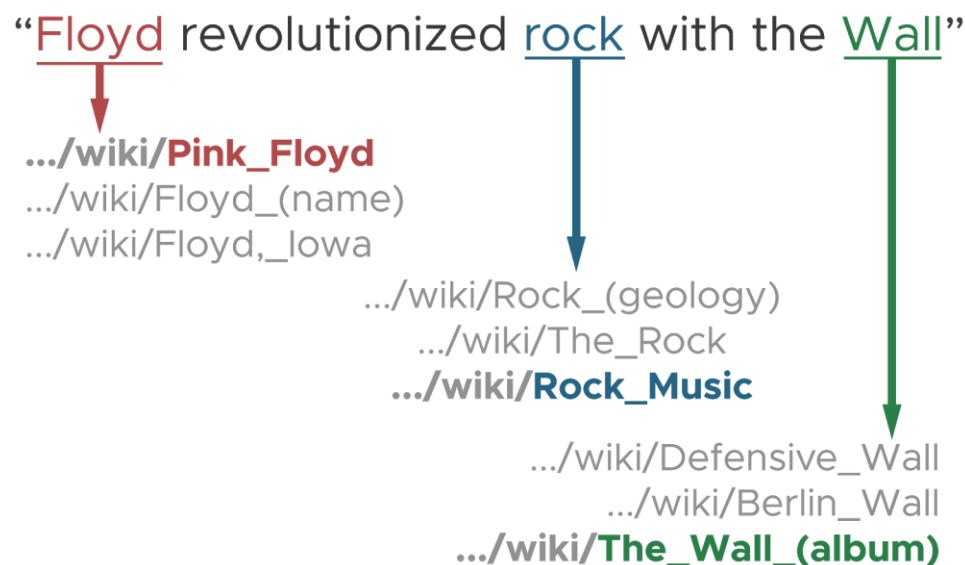
- NED entails connecting an entity mention to a concept in a given ontology
- NER only detects the most likely class
 - it does not link to specific concepts



Icons credit: Creative Mania & Rahmat Dwi Cahyo from the Noun Project

Named Entity Disambiguation

- It is necessary to understand the context of use



<https://blogs.oracle.com/ai-and-datascience/post/named-entity-disambiguation-with-knowledge-graphs> (latest access: June 2021)

Approaches to NER

- Knowledge engineering
 - Rule-based NER
 - Ontology-based NER
- Machine Learning-based NER
 - Standard classifiers
 - E.g., NNs, kNN, SVMs, Bayesian classifiers
 - Sequence models
 - RNNs, LSTMs, Transformers

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State-of-the-art,
but...

Approaches to NER: comparison

- Knowledge engineering
 - High precision
 - Manually generated rules
 - Requires a small amount of training data
 - Strong domain dependence
 - Expensive development, testing, and updating
- Machine Learning-based NER
 - High recall
 - Requires lots of training data
 - No hand-crafted rules
 - Not dependent on semantic models

Rule-based NER

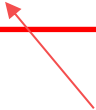
- It relies on regular expressions to identify and classify named entities
- Usually tailored to a specific domain
- E.g., identify drug names in the text
- Improve the generalization power if compared with resource-based systems
- They fail in most open-domain classes

Rule-based NER: example

- Examples of rules suitable for detecting and classifying people:
 - A sequence of words starting with a capital letter, followed by lowercase letters
 - E.g., Barack Obama
 - May contain a prefix title (Dr., Mr., Prof.)
 - E.g., Prof. Enrico Fermi
 - May contain an initial in the middle
 - E.g., George W. Bush
 - May contain a designation indicator prefix
 - E.g., Elon Musk CEO
 - Never include special characters: %/\$/=...

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 - Never include special characters: %/\$/=...



New proper names
constantly emerge!

Rule-based NER

- Example of regular expression: “`^P[a-z][aeiou]`”
 - starts with a capital letter “P”
 - It is the first word on a line
 - The second letter is a lower case letter
 - It is exactly three letters long
 - The third letter is a vowel
 - The regular expression would be where

`^` indicates the beginning of the string

`[a-z]` any letter in range a to z

`[aeiou]` – any vowel

Rule-based NER

- Regular expression list
 - \w (word char) any alpha-numeric
 - \d (digit char) any digit
 - \s (space char) any whitespace
 - . (wildcard) anything
 - \b word boundary
 - ^ beginning of string
 - \$ end of string
 - ? For 0 or 1 occurrences
 - + for 1 or more occurrences
 - specific range of number of occurrences: {min,max}
 - A{1,5} One to five A's.
 - A{5,} Five or more A's
 - A{5} Exactly five A's

Xiang Rneg. ML for Knowledge Extraction and Reasoning. USC Computer Science (CSCI699).

Rule-based NER: example

- Regular expression list for a telephone number
 - blocks of digits separated by hyphens

RegEx = $(\backslash d+\backslash -)^+\backslash d+$

- matches valid phone numbers like 011-909-7099 and 011-011
- incorrectly extracts social security numbers 123-45-6789
- fails to identify numbers like 800.865.1125 and (800)865-CARE

Improved RegEx = $(\backslash d\{3\}[-.\ ()])\{1,2\}[\backslash dA-Z]\{4\}$

Xiang Rneg. ML for Knowledge Extraction and Reasoning. USC Computer Science (CSCI699).

Rule-based NER: example

- Regular expression to extract locations
 - Capitalized word + {city, center, river} indicates location
 - E.g., New York city, Central Park
 - Capitalized word + {street, boulevard, avenue} indicates location
 - E.g., Fifth avenue

Rule-based NER: example

- Capitalization is useful but...
 - First word of a sentence is capitalized by default
 - Titles are often all capitalized
 - Some languages have peculiar capitalization rules
 - E.g., all nouns in German all capitalized
 - Nested entity names contain non-capital words
 - E.g., University of West Virginia

Xiang Rneg. ML for Knowledge Extraction and Reasoning. USC Computer Science (CSCI699).

Rule-based NER: challenges

- The same entity can have multiple variants of the same name
 - E.g., Luca Cagliero, Prof. Cagliero, Professor L. Cagliero
- Proper names can be ambiguous
 - E.g., Felice Buonanno, Santa Pазienza, Guido Di Rado, Immacolata Sforza

Rule-based NER: challenges

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 - E.g., Luca Cagliero, Prof. Cagliero, Professor L. Cagliero
- Proper names can be ambiguous
 - E.g., Felice Buonanno, Santa Pazienza, Guido Di Rado, Immacolata Sforza

Need for ad hoc resolution steps!

Enhanced Rule-based NER

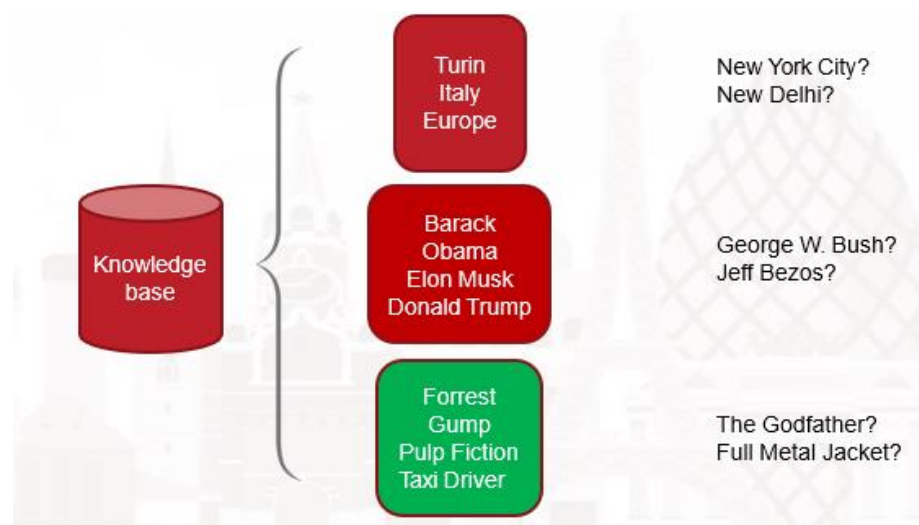
- Combine regular expressions with basic NLP preprocessing steps
 - Part-of-speech (POS) tagging
 - Mark each word as a noun, verb, preposition, etc.
 - Syntactic parsing
 - Identify phrases: NP, VP, PP
 - Text-to-audio
 - Identify phonemes
 - Assign semantic word categories
 - E.g., from the WordNet lexical database (<https://wordnet.princeton.edu/>)
 - KILL: kill, murder, assassinate, strangle, suffocate

Knowledge-based NER

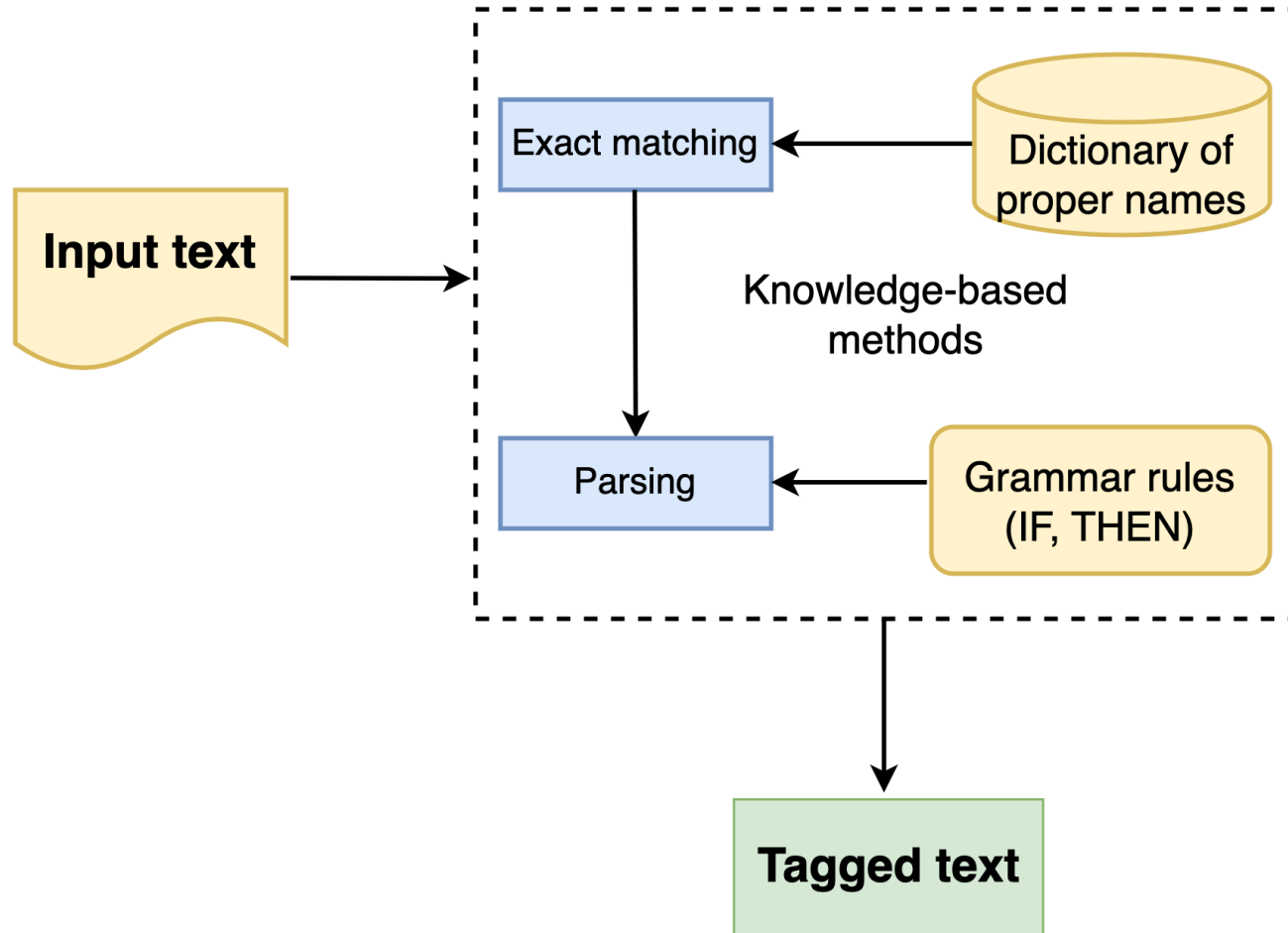
- Use existing domain-specific knowledge graphs or ontologies as resources for discovering specific named entities in text
 - These approaches work well when the ontological resources are exhaustive for the task
 - They fail when some entities are missing



Knowledge-based NER: example



Knowledge-based NER



- Yago
- Dbpedia
- Wikidata
- ...



Knowledge-based NER

- Yago

- Semantic knowledge base, derived from Wikipedia WordNet and GeoNames
- More than 10 million entities
 - persons, organizations, cities
- more than 120 million facts about these entities

Yago: a core of semantic knowledge. FM Suchanek, G Kasneci, G Weikum.

Proceedings of the 16th Proceedings of the 16th international conference on World Wide Web. Pages 697-706 , 2007. ACM.

Knowledge-based NER

- NERD

- set of mappings established manually between the taxonomies of named entity types
- NERD core
 - Thing
 - Amount
 - Animal
 - Event
 - Function
 - Location
 - Organization
 - Person
 - Product
 - Time

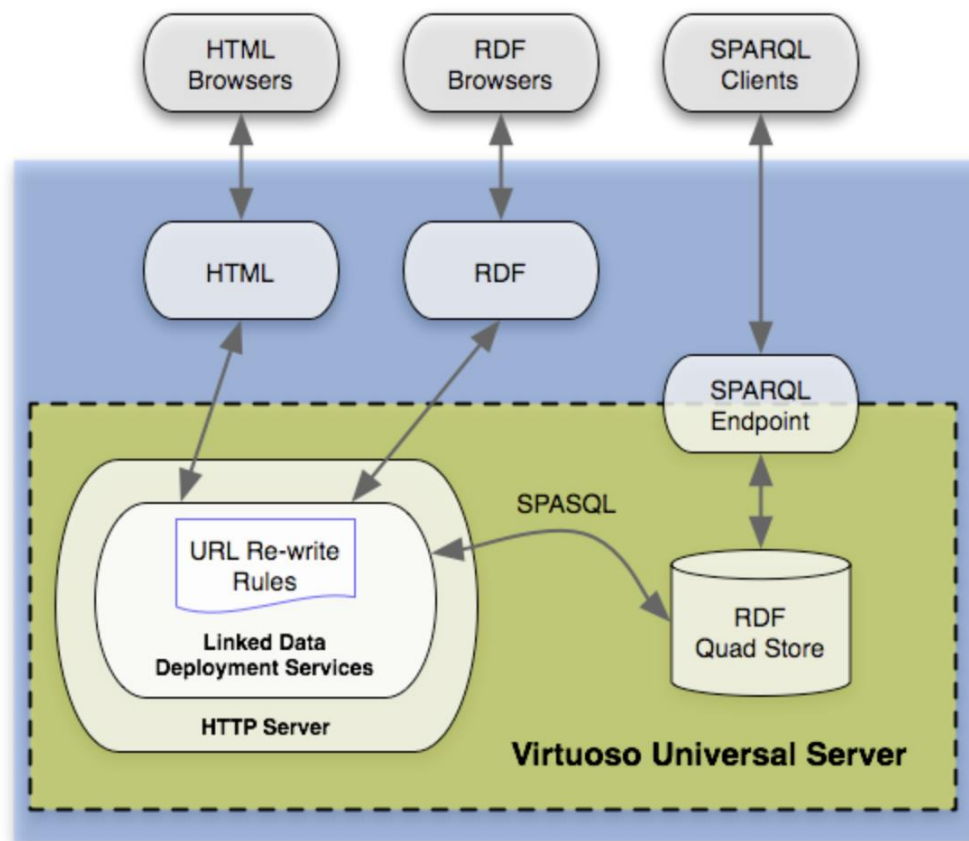
Knowledge-based NER

- DBPedia

- Crowdsourced community effort to extract structured information from Wikipedia
- NERD core
 - 4.58 million things, out of which 4.22 million are classified in a consistent ontology, including
 - 1,445,000 persons
 - 735,000 places (including 478,000 populated places)
 - 411,000 creative works (including 123,000 music albums, 87,000 films)
 - 241,000 organizations (including 58,000 companies and 49,000 educational institutions)
 - 251,000 species
 - 6,000 diseases

Knowledge-based NER

- DBPedia

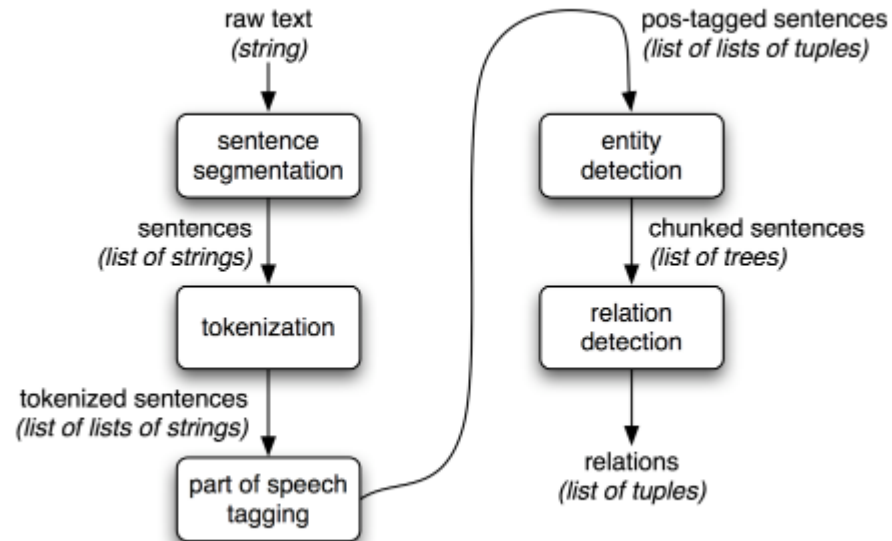


Knowledge-based NER

- Natural Language Toolkit (NLTK)
 - leading platform for building Python programs to work with human language data
 - easy-to-use interfaces to over 50 corpora and lexical resources such as WordNet
 - a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries, and an active discussion forum.

Knowledge-based NER

- Natural Language Toolkit (NLTK)

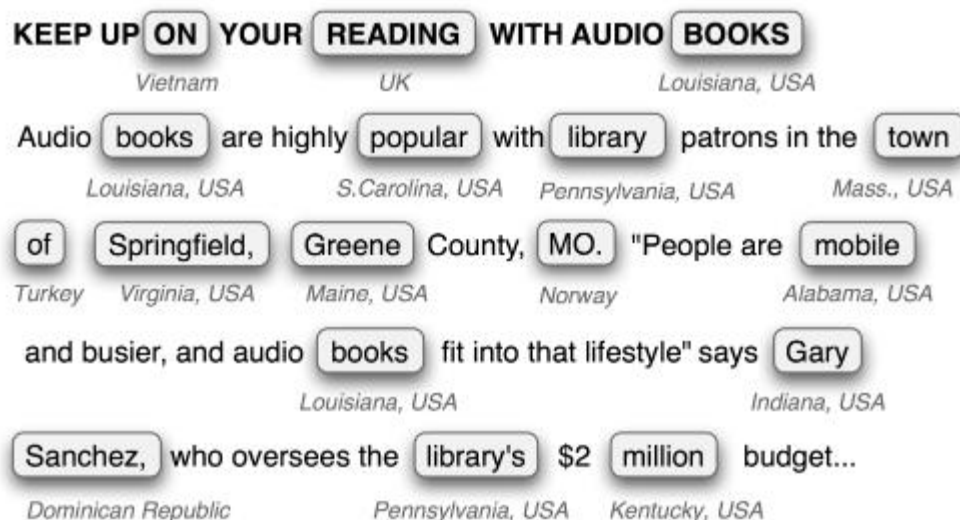


<http://www.nltk.org/book/ch07.html> (latest access: June 2021)

Knowledge-based NER

- NER using NLTK

- It supports the automatic recognition of named entities
 - E.g., Organizations, persons, ...
- The extraction of relations between them
 - E.g., looking for relations between specified types of named entity (person with the corresponding organization)



Knowledge-based NER

- Wikidata

- free, open, multilingual knowledge base
- It provides end-users with access to Wikipedia
- Wikidata content can be read and edited by both humans and machines
- Central storage repository
 - Consists of items, each one having an identifier (Q), a label, a description, any number of aliases
- To access Wikidata
 - Wikidata query and Reasonator tools
 - Lua Scribunto interface
 - **Wikidata API**

<https://wikidata.org> (latest access: June 2021)

Knowledge-based NER

- Wikidata

The image shows a Wikidata entry for Douglas Adams (Q42) with various annotations. The entry includes a label, a description, and a list of statements. The 'educated at' statement is highlighted with a green box, and its details are shown in a blue box. The '2 references' section is highlighted with a red box, and its details are shown in a red box. The 'Brentwood School' statement is highlighted with a red box, and its details are shown in a red box. The '0 references' section is highlighted with a red box.

label — Douglas Adams (Q42) — item identifier

description — English writer and humorist
Douglas Noël Adams | Douglas Noel Adams — aliases
► In more languages

Statements

property — educated at — value — St John's College

rank —

statement group —

qualifiers —

opened references —

collapsed reference —

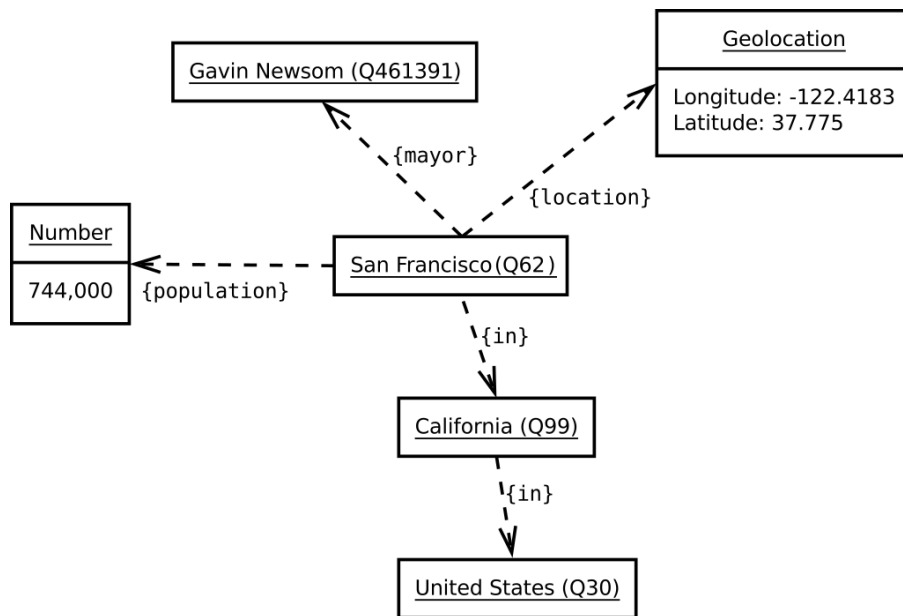
Statement	Value	Qualifiers
educated at	St John's College	end time: 1974, academic major: English literature, academic degree: Bachelor of Arts, start time: 1971
2 references	Encyclopædia Britannica Online	reference URL: http://www.nndb.com/people/731/000023662/, original language of work: English, retrieved: 7 December 2013, publisher: NNDB, title: Douglas Adams (English)
Brentwood School		end time: 1970, start time: 1959
0 references		

<https://wikidata.org> (latest access: June 2021)

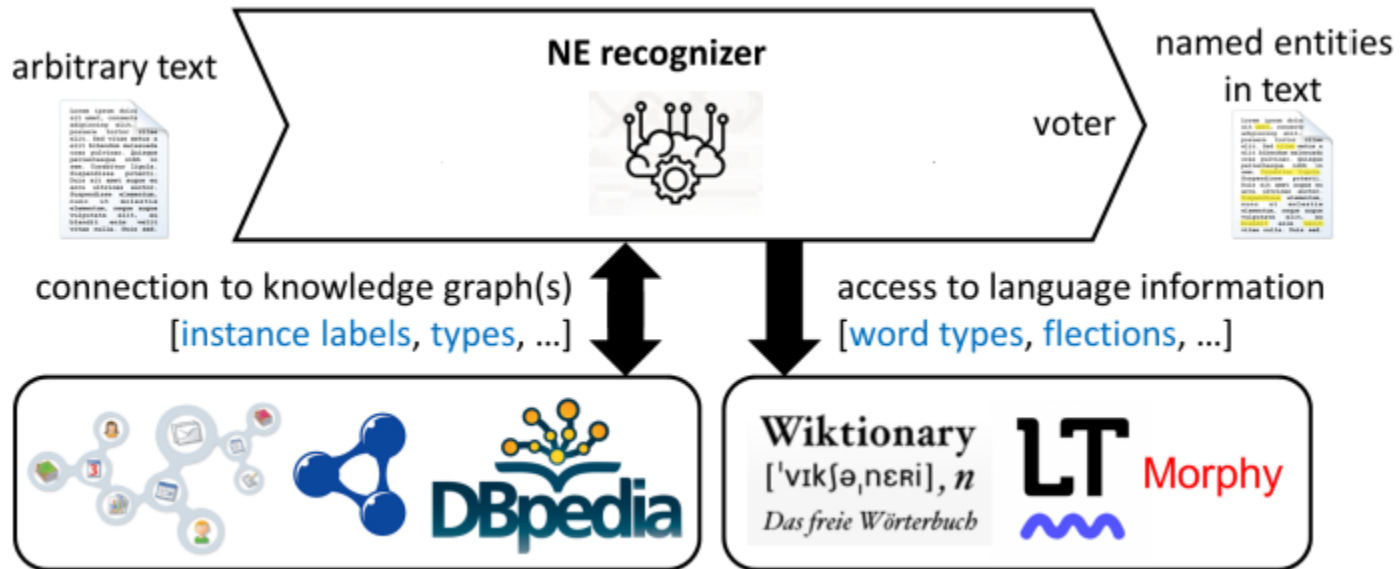
Knowledge-based NER

- Wikidata

- Consists of items, each one having an identifier (Qxy), a label, a description, any number of aliases
- A statement describes a detailed characteristic of an item
 - It consists of a property and a value
 - A property can link an item to an external database

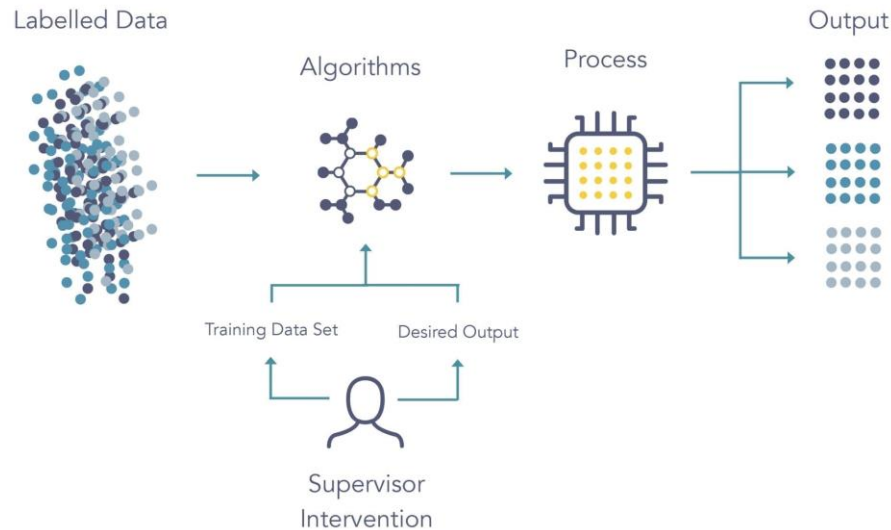


Knowledge-based NER example



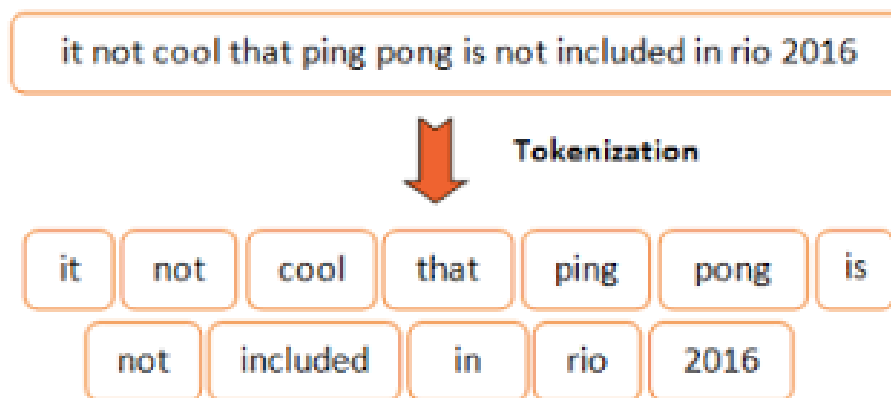
Machine Learning-based NER

- Apply a Machine Learning pipeline to leverage human-generated annotations
 - Learn a predictive model from annotated data
 - Apply to model to unlabeled data



Machine Learning-based NER

1) Tokenization



Machine Learning-based NER

2) Human annotation

Intesa
Sanpaolo
S.p.A.
Or
simply
ISP
Group
...

Human
Annotation



Intesa	B-ORG
Sanpaolo	I-ORG
S.p.A.	I-ORG
or	O
simply	O
ISP	B-ORG
Group	I-ORG
...	...

Machine Learning-based NER

- Named Entity Detection
 - B-prefix indicates the beginning of a named entity
 - I-prefix refers to a word inside the named entity
 - O-prefix refers to a word outside the named entity
- Named Entity Classification
 - Person names
 - Organizations
 - E.g., companies, governmental organizations
 - Locations
 - Miscellaneous
 - E.g., sport events

Machine Learning-based NER

3) Feature engineering

Intesa
Sanpaolo
S.p.A.
Or
simply
ISP
Group
...

Feature
extraction



Word	POS	Symb. Count
Intesa	NOUN	0
Sanpaolo	NOUN	0
S.p.A.	NOUN	2
Or	CONJ	0
simply	ADV	0
ISP	NOUN	0
Group	NOUN	0
...	...	...

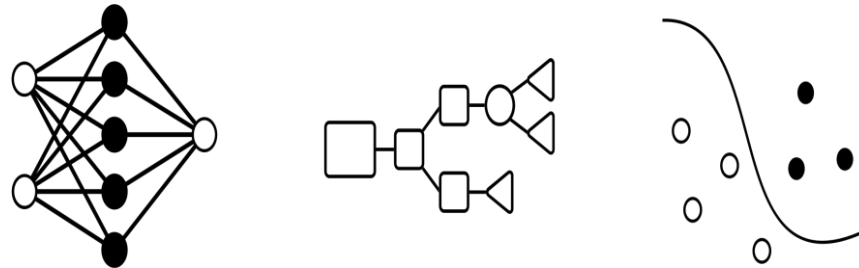
Machine Learning-based NER

Training dataset

Word	POS	Symb. Count	Class
Assicurazioni	NOUN	0	B-ORG
Generali	NOUN	0	I-ORG
S.p.A.	NOUN	2	I-ORG
or	CONJ	0	O
simply	ADV	0	O
Generali	NOUN	0	B-ORG
Group	NOUN	0	I-ORG

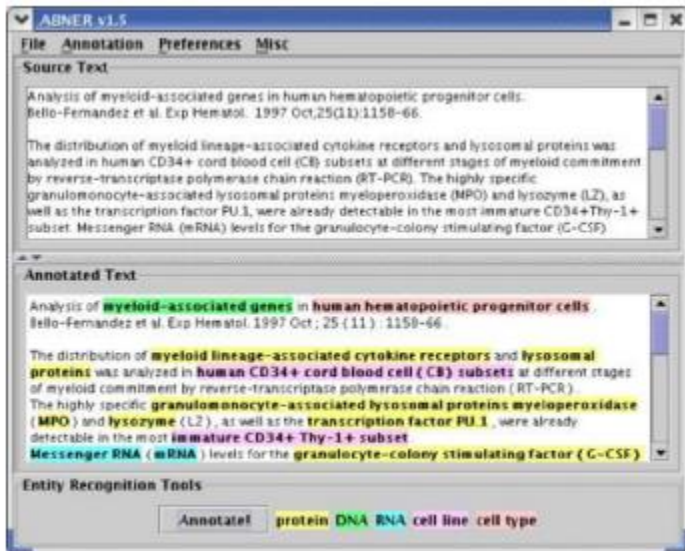
Machine Learning-based NER

4) Classifier training

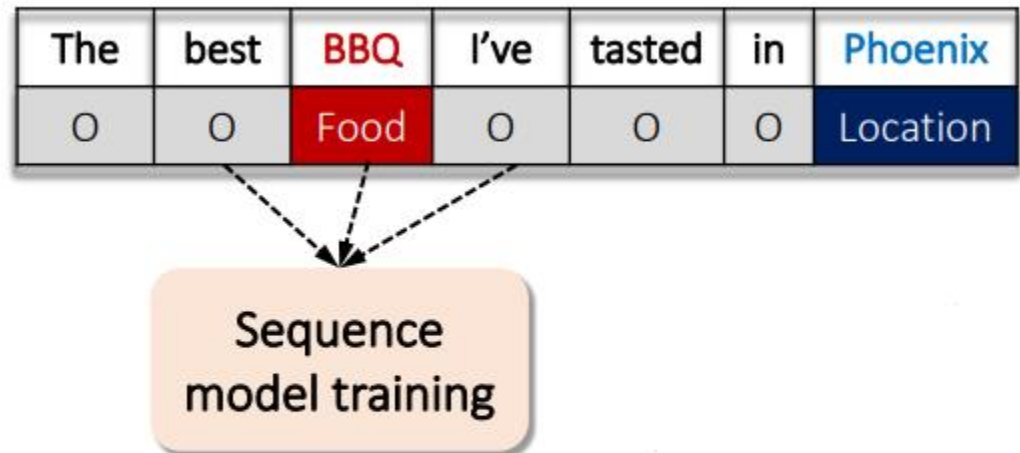


Sequence labeling for NER

- Classify using sequential models
 - E.g., RNNs, LSTMs

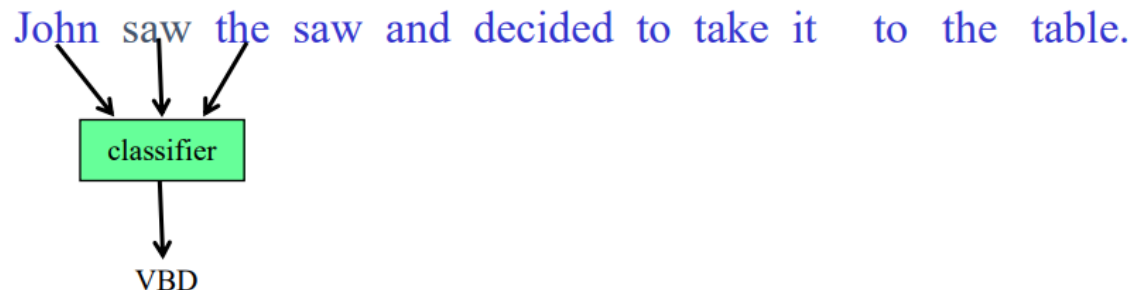


A manual annotation interface



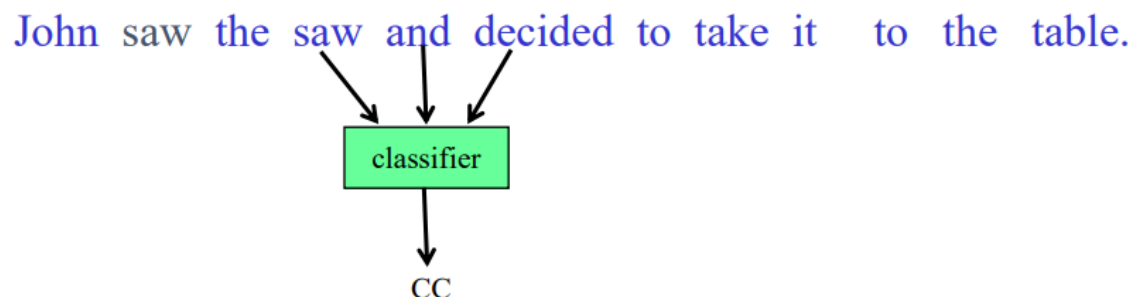
Sequence labeling for NER

- Classify each token independently but...
 - Use as input features the information about the surrounding tokens
 - Sliding window approach



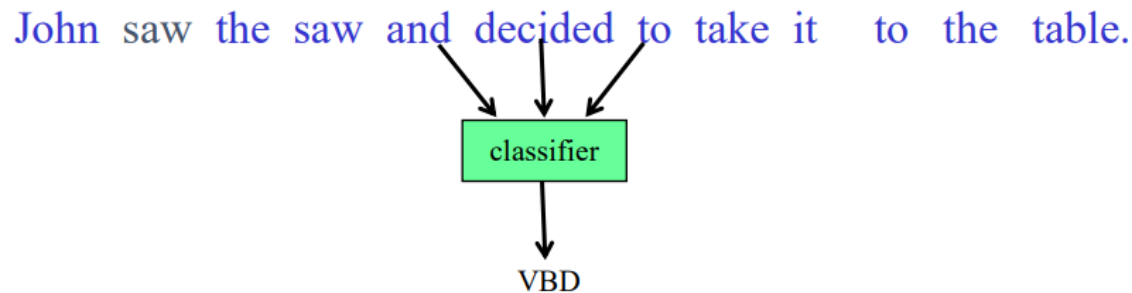
Sequence labeling for NER

- Classify each token independently but...
 - Use as input features the information about the surrounding tokens
 - Sliding window approach



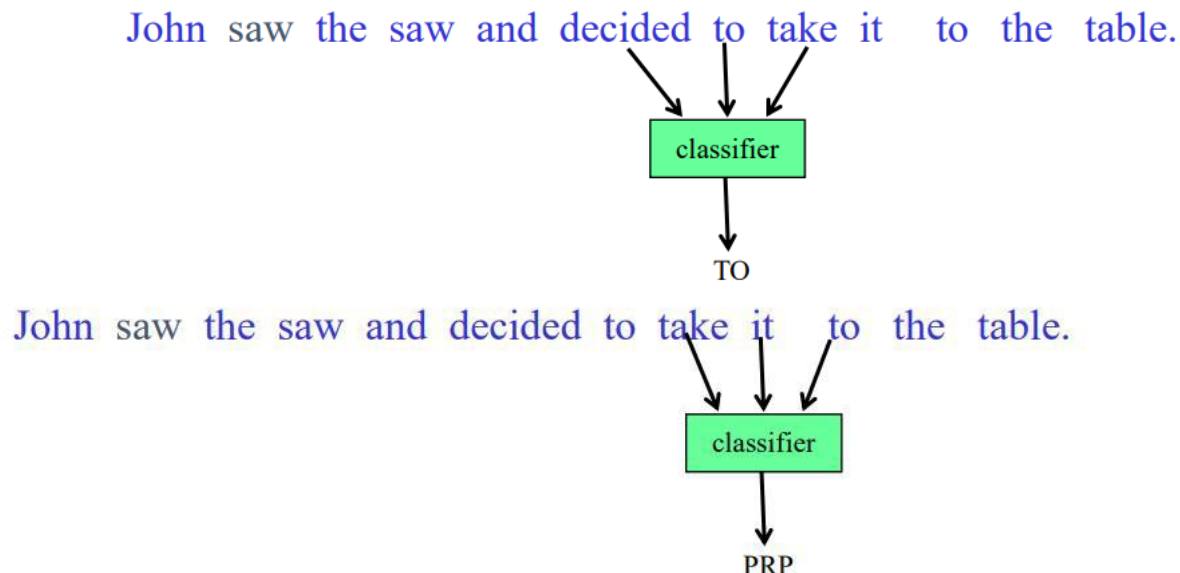
Sequence labeling for NER

- Classify each token independently but...
 - Use as input features the information about the surrounding tokens
 - Sliding window approach



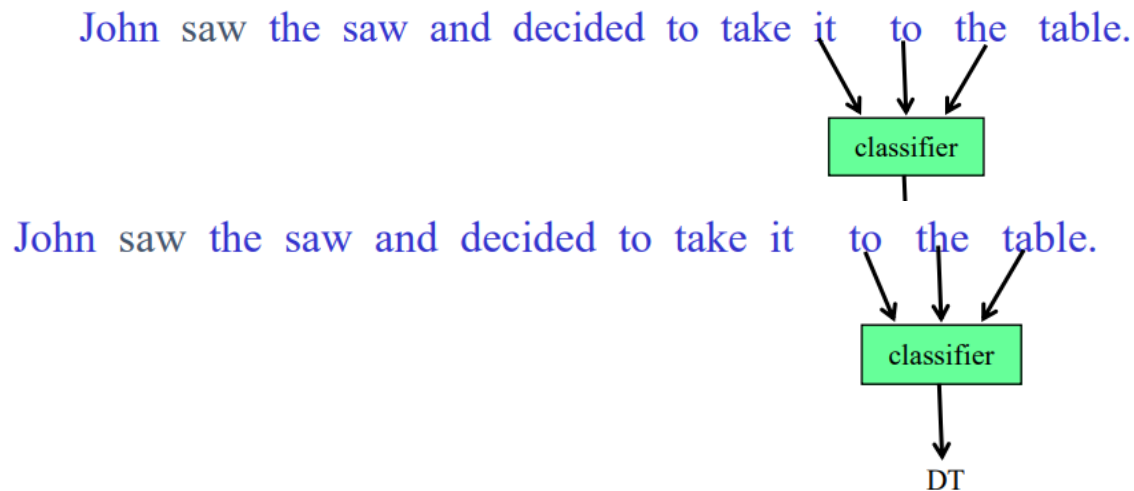
Sequence labeling for NER

- Classify each token independently but...
 - Use as input features the information about the surrounding tokens
 - Sliding window approach

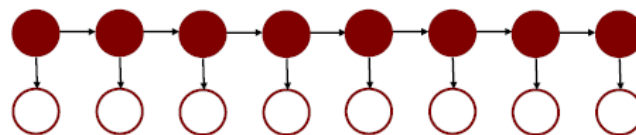


Sequence labeling for NER

- Classify each token independently but...
 - Use as input features the information about the surrounding tokens
 - Sliding window approach

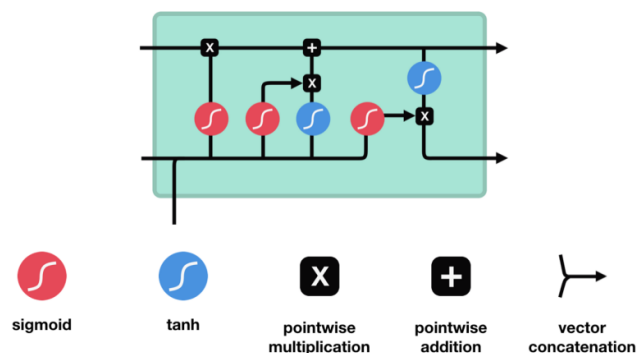


Sequence labeling: techniques

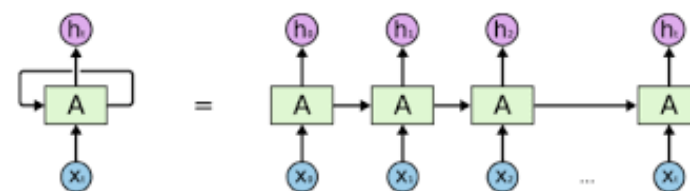


Hidden Markov Models

Conditional Random Forests



LSTMs

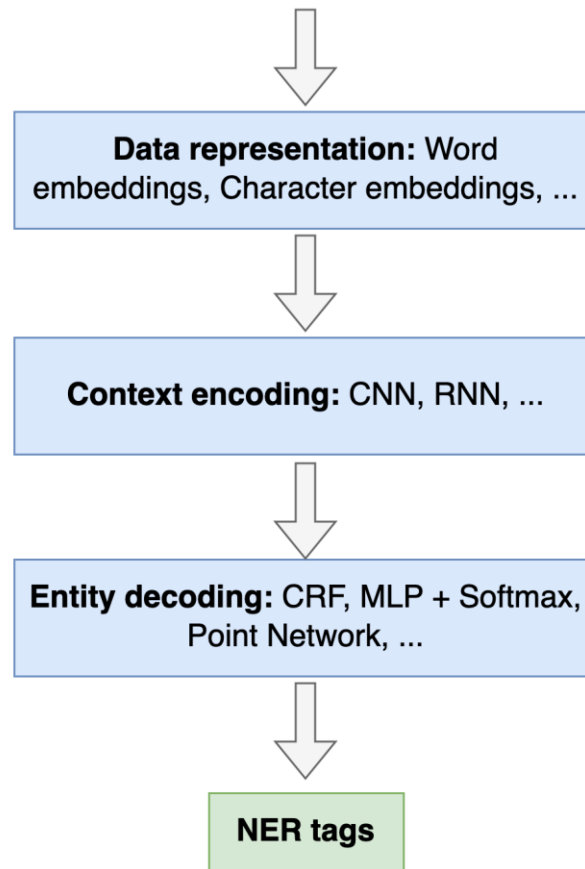


An unrolled recurrent neural network.

Recurrent Neural Networks

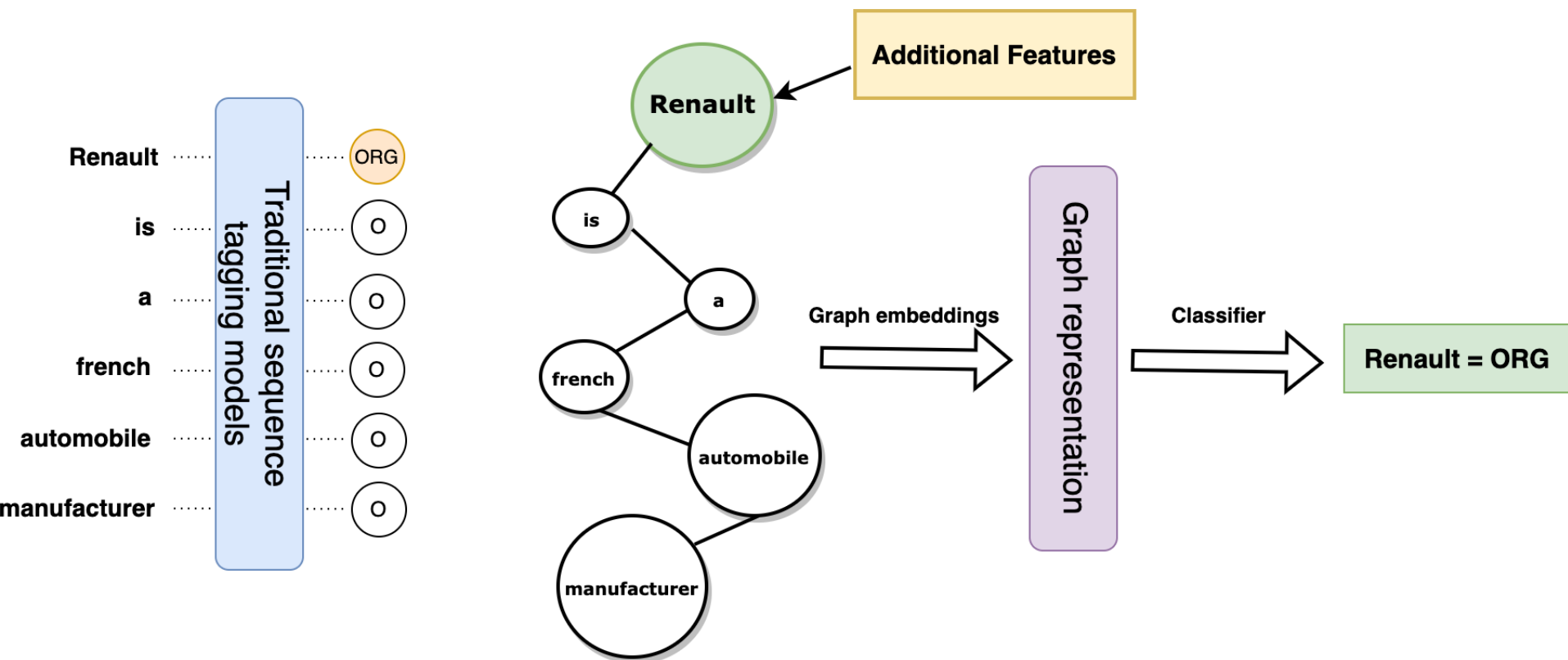
Deep Learning-based NER

Learn to walk before you run.



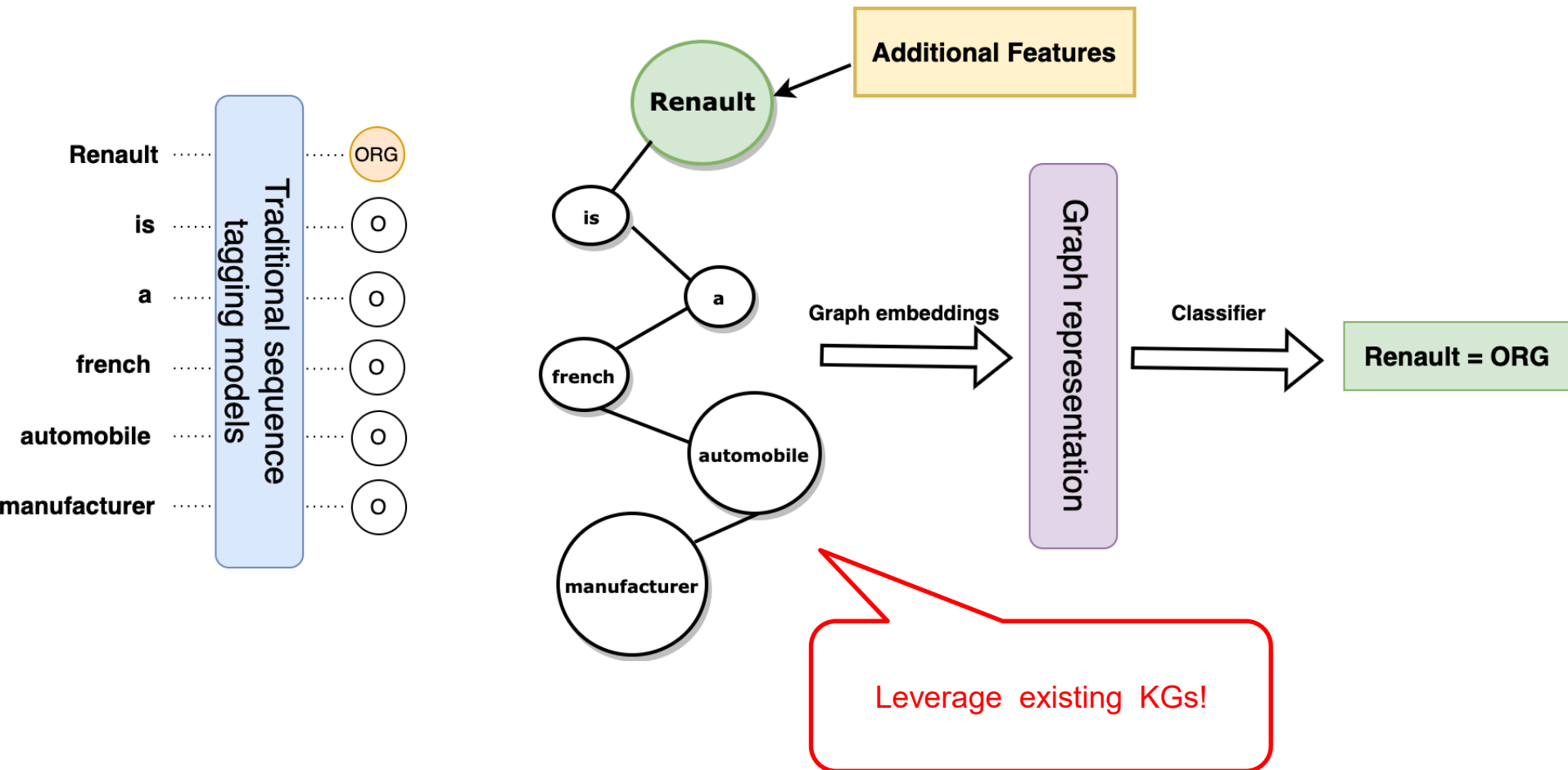
<https://arxiv.org/html/2401.10825v1>

Deep Learning-based NER

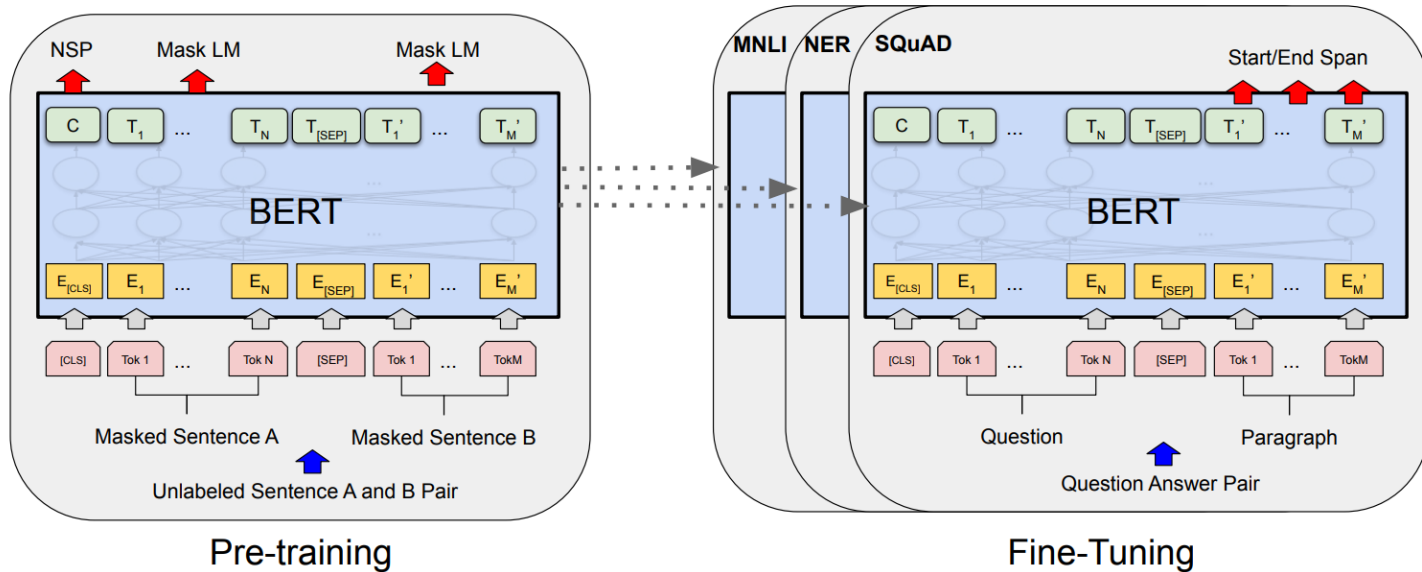


<https://arxiv.org/html/2401.10825v1>

Deep Learning-based NER

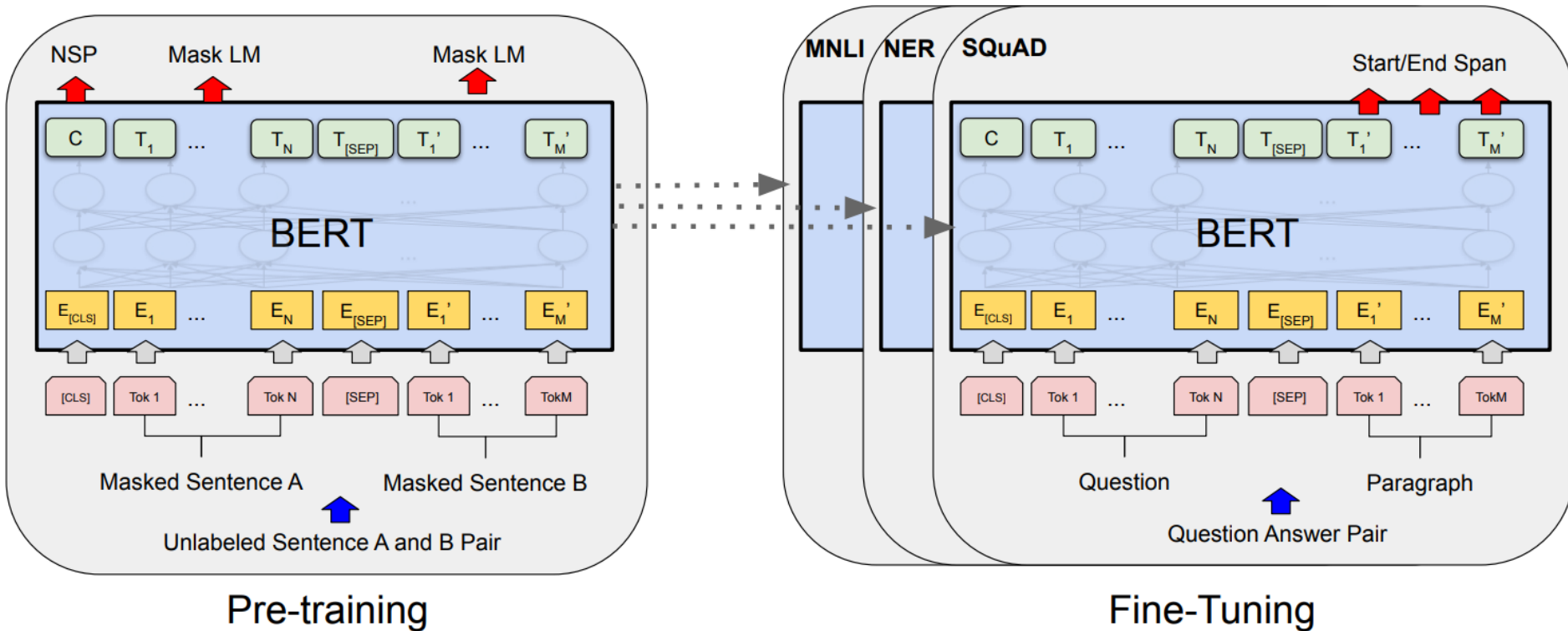


Transformers for NER



Transformers for NER

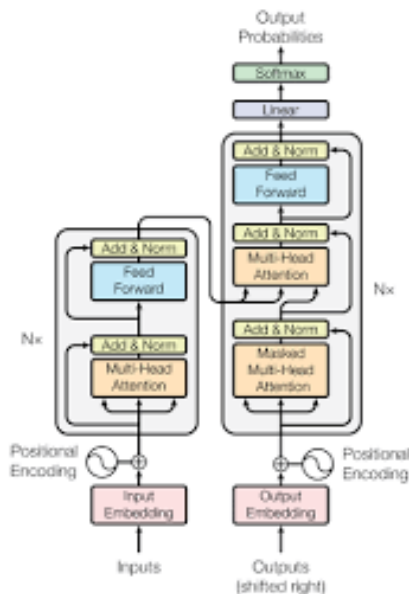
- Leverage attention-based architecture
 - Fine-tune the model for the NER task



BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova. Proceedings of NAACL-HLT 2019.

Transformers for NER: the LUKE architecture

- **L**anguage **U**nderstanding with **K**nowledge-Based **E**mbeddings
 - Presented at EMNLP'20
 - State-of-the-art approach for NER
- Key idea
 - use the attention mechanism to attend both word and entity relationships
- Based on a pretrained BERT Masked Language Model



Ikuya Yamada, Akari Asai, Hiroyuki Shindo, Hideaki Takeda, Yuji Matsumoto:

LUKE: Deep Contextualized Entity Representations with Entity-aware Self-attention. EMNLP (1) 2020: 6442-6454

Transformers for NER: the LUKE architecture

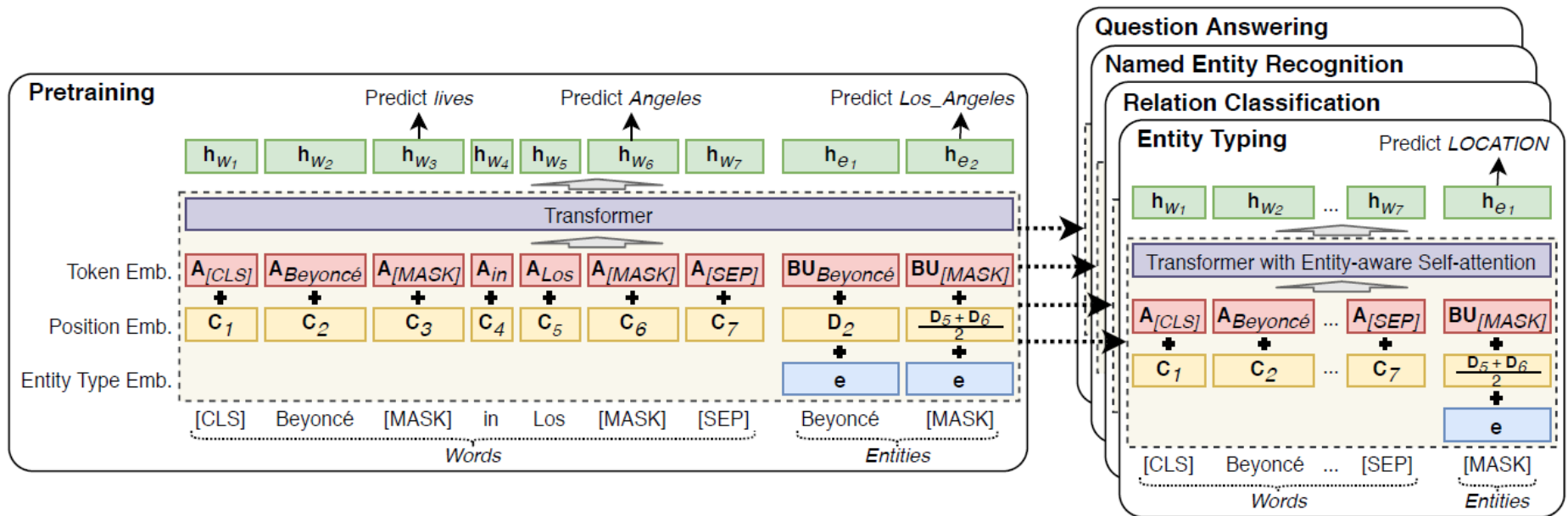
- Pretrained contextualized representations of both words and entities
 - Trained on large-scale entity-annotated corpus obtained from Wikipedia
 - It treats hyperlinks in Wikipedia as entity annotations



Ikuya Yamada, Akari Asai, Hiroyuki Shindo, Hideaki Takeda, Yuji Matsumoto:

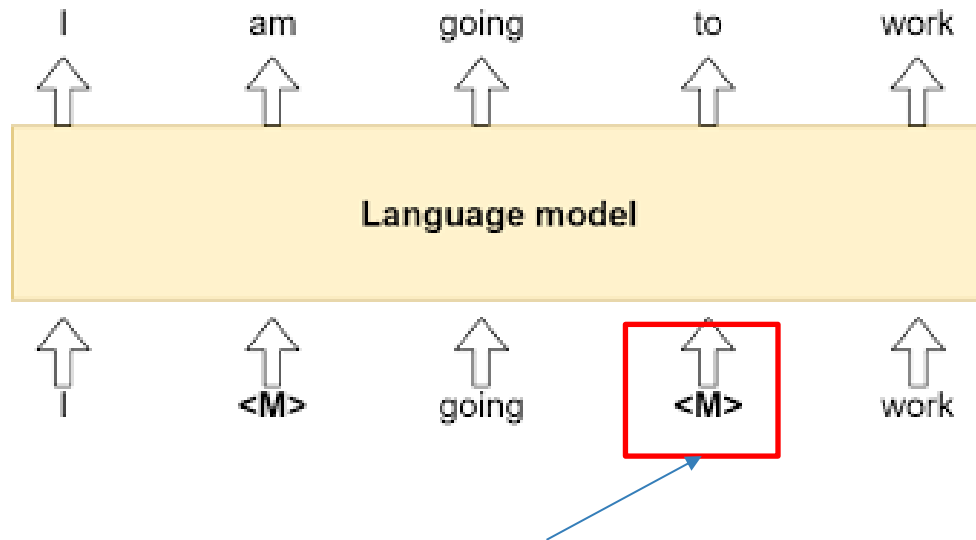
LUKE: Deep Contextualized Entity Representations with Entity-aware Self-attention. EMNLP (1) 2020: 6442-6454

Transformers for NER: the LUKE architecture



Transformers for NER: the LUKE architecture

- Extended Masked Language Model
 - Entities within the annotated text are randomly masked
 - The trained model predicts the original masked entities



You can mask entities as well

Transformers for NER: the LUKE architecture

- Entities are treated as independent tokens
- LUKE computes the intermediate and output representations of both word and entity tokens
 - It inherently models the relationships between entities
- Input embeddings
 - Token embedding
 - Representation of the token (either word or entity)
 - Position embedding
 - Position of the token in the word sequence
 - If an entity name contains multiple words, the single embeddings are averaged
 - Entity type embedding
 - Whether the token is an entity or not

Transformers for NER: the LUKE architecture

- Entity-aware attention mechanism
 - It relates tokens (either entities or words) each other based on the attention score between each pair of token
- Input vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$
- Output vectors $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k$

$$\mathbf{y}_i = \sum_{j=1}^k \alpha_{ij} \mathbf{V} \mathbf{x}_j$$
$$e_{ij} = \frac{\mathbf{K} \mathbf{x}_j^\top \mathbf{Q} \mathbf{x}_i}{\sqrt{L}}$$
$$\alpha_{ij} = \text{softmax}(e_{ij})$$
$$e_{ij} = \begin{cases} \mathbf{K} \mathbf{x}_j^\top \mathbf{Q} \mathbf{x}_i, & \text{if both } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are words} \\ \mathbf{K} \mathbf{x}_j^\top \mathbf{Q}_{w2e} \mathbf{x}_i, & \text{if } \mathbf{x}_i \text{ is word and } \mathbf{x}_j \text{ is entity} \\ \mathbf{K} \mathbf{x}_j^\top \mathbf{Q}_{e2w} \mathbf{x}_i, & \text{if } \mathbf{x}_i \text{ is entity and } \mathbf{x}_j \text{ is word} \\ \mathbf{K} \mathbf{x}_j^\top \mathbf{Q}_{e2e} \mathbf{x}_i, & \text{if both } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are entities} \end{cases}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{B} \mathbf{T} \mathbf{m} + \mathbf{b}_o)$$

$$\mathbf{m} = \text{layer_norm}(\text{gelu}(\mathbf{W}_h \mathbf{h}_e + \mathbf{b}_h))$$

Additional reading on LUKE



- Ikuya Yamada, Akari Asai, Hiroyuki Shindo, Hideaki Takeda, Yuji Matsumoto. LUKE: Deep Contextualized Entity Representations with Entity-aware Self-attention. EMNLP (1) 2020: 6442-6454
- Please read the paper: <https://arxiv.org/pdf/2010.01057.pdf>

Prompt-based NER

- Prompting LLMs with ad hoc queries and examples
 - Adopt pretrained models
 - Fine-tune pretrained LLMs on the NER task

Prompt-based NER



You

Prompt: Please identify the named entities in the sentence provided, classifying them as either "CHM" for chemical entities or "DIS" for disease entities. "CHM" refers to chemical substances, including drugs, compounds, biomolecules, and more, like "Aspirin" or "Carbon Dioxide". "DIS" pertains to medical conditions or diseases, such as "Diabetes" or "Glioblastoma". Present your findings in JSON format, with keys "CHM" and "DIS", listing the identified entities under the appropriate category.

Sentence for analysis: "In this model of chronic renal failure."



ChatGPT

Based on the provided sentence, here are the identified named entities classified as "CHM" and "DIS" in JSON format:

```
json Copy code
{
  "CHM": [],
  "DIS": ["chronic renal failure"]
}
```

<https://arxiv.org/html/2401.10825v1>

Low-Resource NER

- The use of Neural Networks requires a large amount of training data
 - Costly
 - Difficult to obtain
 - Biased
- Possible solutions
 - Zero-Shot Learning
 - Few-Shot Learning
 - Transfer Learning
 - Data Augmentation
 - Active Learning

Zero- and Few-Shot Learning for NER

- Zero-Shot

- Use a pre-trained model to assign classes to elements that the model has never encountered before
 - In NER it is a way of detecting new types of named entities
 - Common use of pretrained LLMs in absence of training data

- Few-Shot

- Build accurate models with less training data
 - Apply transformations to the data or changes in the algorithms
 - Suited to LLMs with in-context learning

Transfer Learning for NER

- Transfer Learning
 - Take knowledge that has been obtained while solving one task
 - Apply it to a related task
- Transfer Learning using Deep Networks
 - Pretrain a model on a large amount of general text data
 - Then fine-tune it on a smaller dataset specifically for the target NER task
 - Particularly effective with Transformers

Data Augmentation for NER

- Data Augmentation

- Artificially increase the amount of training data by creating modified copies of a dataset using existing data.
 - Make small changes to the original data
 - Use generative models
 - Adopting paraphrasing (Seq2Seq) models

- Caveat

- Changes to words or sentences might affect token classification performance significantly
- High risk of introducing bias or errors

Active Learning for NER

- Active Learning
 - Semi-supervised Learning
 - If a learner is able to choose the data that it wishes to learn from then it can perform better otherwise he should use traditional learning schemes
- Open issues
 - What constitutes the most informative data?
 - How can the learner recognize this data?
- Possible approach
 - Uncertainty sampling: the model selects the examples for which its current prediction is the least reliable

Performance metrics for NER

- Exact evaluation

- **Precision:** proportion of named entities successfully recognized by a model in relation to the total number of named entities
- **Recall:** number of relevant named entities retrieved by a model against the total number of relevant named entities
- **F1-score:** model's ability to effectively detect named entities, with a trade-off between precision and recall (harmonic mean)

- How to handle multi-class

- **Macro-Average:** the metric (for instance, F1-score) is computed for each class separately, with the macro-average corresponding to the mean of these values
- **Micro-average:** it gives equal weight to each sample

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Performance metrics for NER

- Relaxed evaluation

- Go beyond a strict classification and consider only partial matching
- Common strategies
 - Message Understanding Conference
 - <https://github.com/jantrienes/nereval>
 - Semantic Evaluation Workshop
 - <https://semeval.github.io/>

Performance metrics for NER

- Message Understanding Conference
 - Compare the responses with a Ground Truth
 - Categories of matches:
 - **Correct** (COR): both are the same
 - **Incorrect** (INC): the output of a system and the golden annotation don't match
 - **Partial** (PAR): system and the golden annotation are somewhat “similar” but not the same
 - **Missing** (MIS): a golden annotation is not captured by a system
 - **Spurious** (SPU): system produces a response which doesn't exist in the golden annotation
 - Implementation
 - <https://github.com/jantrienes/nereval>

Performance metrics for NER

- SemEval Workshops

- four different ways to measure precision/recall/F1-score results based on the metrics defined by MUC
 - **Strict**: exact boundary surface string match and entity type;
 - **Exact**: exact boundary match over the surface string, regardless of the type;
 - **Partial**: partial boundary match over the surface string, regardless of the type;
 - **Type**: some overlap between the system tagged entity and the gold annotation is required;
- Implementation
 - <https://semeval.github.io/>

Performance metrics for NER

- SemEval Workshops

Exact Match (i.e., *strict* and *exact*)

$$\text{Precision} = \frac{COR}{ACT} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{COR}{POS} = \frac{TP}{TP + FN}$$

Partial Match (i.e., *partial* and *type*)

$$\text{Precision} = \frac{COR + 0.5 \times PAR}{ACT} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{COR + 0.5 \times PAR}{POS} = \frac{COR}{ACT} = \frac{TP}{TP + FP}$$

Performance metrics for NER

- Mapping between MUC and SemEval scores

Scenario	Golden Standard		System Prediction		Evaluation Schema			
	Entity Type	Surface String	Entity Type	Surface String	Type	Partial	Exact	Strict
III	brand	TIKOSYN			MIS	MIS	MIS	MIS
II			brand	healthy	SPU	SPU	SPU	SPU
V	drug	warfarin	drug	of warfarin	COR	PAR	INC	INC
IV	drug	propranolol	brand	propranolol	INC	COR	COR	INC
I	drug	phenytoin	drug	phenytoin	COR	COR	COR	COR
I	Drug	theophylline	drug	theophylline	COR	COR	COR	COR
VI	group	contraceptives	drug	oral contraceptives	INC	PAR	INC	INC

Performance metrics for NER

- Mapping between MUC and SemEval scores

Measure	Type	Partial	Exact	Strict
Correct	3	3	3	2
Incorrect	2	0	2	3
Partial	0	2	0	0
Missed	1	1	1	1
Spurious	1	1	1	1
Precision	0.5	0.66	0.5	0.33
Recall	0.5	0.66	0.5	0.33
F1	0.5	0.66	0.5	0.33

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- Attribution + Noncommercial + NoDerivatives



- Affiliation

- The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!